

I-CURE-1: A Phase II, single arm study of Pembrolizumab combined with carboplatin in patients with circulating tumor cells (CTCs) positive Her-2 negative metastatic breast cancer (MBC)

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Study Intervention(s): pembrolizumab (Keytruda®)

IND Number: 133923

IND Holder: Lisa E. Flaum, MD

Funding Source: Merck Sharp & Dohme Corp.

Sponsor: Northwestern University

Version Date: 11.30.2021
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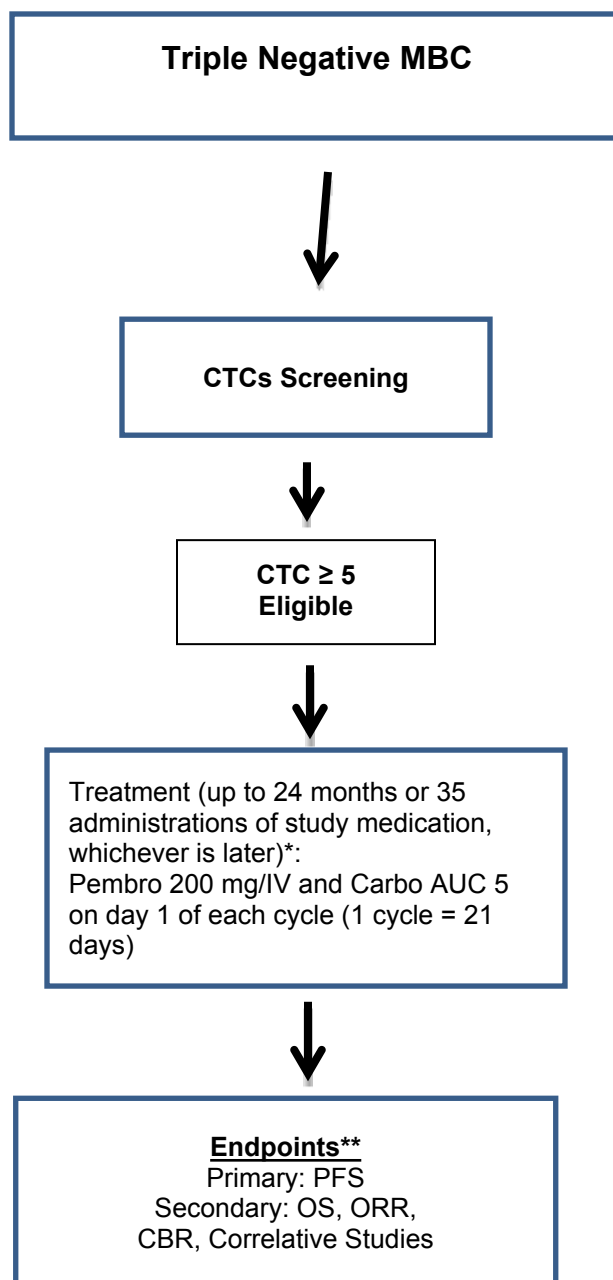
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LIST OF ABBREVIATIONS

AE	Adverse Event
ALT	Alanine Aminotransferase
ALC	Absolute Lymphocyte Count
AST	Aspartate Aminotransferase
BUN	Blood Urea Nitrogen
CBC	Complete Blood Count
CBR	Clinical Benefit Rate
CMP	Comprehensive Metabolic Panel
CR	Complete Response
CTCs	Circulating tumor cells
CTCAE	Common Terminology Criteria for Adverse Events
DLT	Dose Limiting Toxicity
DSMB	Data and Safety Monitoring Board
ECOG	Eastern Cooperative Oncology Group
H&PE	History & Physical Exam
HR	Hormone Receptor (Estrogen or Progesterone Receptors)
IV (or iv)	Intravenously
MBC	Metastatic breast cancer
MTD	Maximum Tolerated Dose
NCI	National Cancer Institute
ORR	Overall Response Rate or Objective Response Rate
OS	Overall Survival
PBMCs	Peripheral Blood Mononuclear Cells
PD	Progressive Disease
PFS	Progression Free Survival
PR	Partial Response
QAM	Quality Assurance Monitor
SAE	Serious Adverse Event
SD	Stable Disease
SGOT	Serum Glutamic Oxaloacetic Transaminase
SPGT	Serum Glutamic Pyruvic Transaminase
TILs	Tumor-Infiltrating Lymphocytes
TNBC	Triple negative breast cancer

STUDY SCHEMA



*Patients who achieve CR may stop treatment before 24 months or 35 administrations of study medication, whichever is later

**Safety follow up visit within 30 days of the last treatment dose or before initiation of new anti-cancer treatment; Follow up visits every 9 weeks for 1 year, after 1 year every 12 weeks; Survivor follow up by telephone every 12 weeks until death, withdrawal of consent, or end of study.

STUDY SUMMARY

Title	I-CURE-1: A Phase II, single arm study of Pembrolizumab combined with carboplatin in patients with circulating tumor cells (CTCs) positive Her-2 negative metastatic breast cancer (MBC)
Protocol Date	11.30.2021 (Amendment 6)
Study Duration	60 months
Study Center(s)	Robert H. Lurie Comprehensive Cancer Center Northwestern University, Chicago
Objectives	Evaluate the impact on progression-free survival (PFS) with the combination carboplatin - pembrolizumab in patients with CTC positive, triple negative (TN) MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV
Number of Subjects	up to 40 patients (32 evaluable)
Diagnosis and Main Inclusion Criteria	- Metastatic TNBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de novo stage IV - CTC positive (defined as CTC$\geq$5)
Study design	This is a prospective, single arm, Phase II combination study
Treatment Plan	Patients will be treated with pembrolizumab 200mg intravenously (IV) and carboplatin AUC 5 IV on day 1 of each cycle (1 cycle = 21 days). Treatments continue until patient completes 24 months of treatment or 35 administrations of study medication, whichever is later, or until progressive disease or unacceptable toxicity. Patients will return for a follow up visit within 30 days of last treatment dose and will be followed and assessed every 9 weeks for 1 year and after 1 year every 12 weeks to monitor disease status. Patients will be contacted for survival status every 12 weeks until death, withdrawal of consent, or the end of study, whichever occurs first.
Statistical methods	The trial is a Phase II, prospective study testing the combination of pembrolizumab and carboplatin in patients with HER-2 negative, CTC positive (defined as CTCs $\geq$ 5) MBC. The primary hypothesis to be tested is that the combination of Carboplatin intravenously and pembrolizumab will increase PFS in patients with $\geq$ 5 CTCs compared with the historical median of 4.9 months [1]. The distribution of PFS will be estimated using the Kaplan-Meier (KM) method. Median survival and an associated 95% confidence interval will be estimated from the KM analysis. The null hypothesis of no improvement in survival will be rejected if the lower bound of the confidence interval for median survival is greater than 4.9.

1 BACKGROUND & RATIONALE

1.1 Circulating tumor cells (CTCs) in Metastatic breast Cancer.

Breast cancer mortality has been decreasing considerably over the past two decades as a result of earlier diagnosis and major treatment advances in the adjuvant and metastatic settings [2]. Despite this progress, metastatic disease is still largely considered an incurable condition and 5-year survival rates are less than 25% [2]. Thus, the major treatment objectives in the advanced stage disease remain palliation of symptoms and improvement of quality of life. Importantly, metastatic breast cancer (MBC) is a heterogeneous disease and long-term patient outcome can be influenced by various biological features, as well as by the extent and site of metastatic disease. Typically, widespread visceral disease is associated with symptomatic condition leading to deterioration of the performance status and short survival [3]. Various molecular markers and blood-based tests have been investigated as surrogate for more aggressive disease, among them the most reliable appears enumeration of CTCs [4, 5].

During the last decade, several techniques capable of detecting and quantifying CTCs in cancer patients have been developed. It has been proposed that subpopulation of CTCs with tumor initiating potential act as a central mediator of metastatic dissemination, giving rise to the formation of distant micrometastases, which subsequently generate overt detectable and frequently measurable lesions [6]. In support of this theory, multiple studies have shown that a number of CTCs higher or equal than 5 per 7.5 mL of blood, evaluated before starting systemic treatment, is associated with poor outcome in patients with MBC [7, 8]. In addition, high CTC counts are associated with greater metastatic tumor burden, expressed as number of metastatic sites [8, 9]. Importantly, despite this association, the prognostic value of CTCs is independent from the initial number of metastatic sites [8, 9]. Moreover, a recent pooled analysis of 1944 patients with MBC was performed confirming those data [10]. Moreover, the authors created a clinicopathological prognostic model to determine the added impact of CTCs for progression free survival (PFS) and overall survival (OS). They found that adding CTC count (≤ 5 or ≥ 5) to their predictive model significantly increased the prognostication for OS and PFS. This is in contrast to serum tumor markers, CEA and CA15-3, when checked at 3-5 weeks or 6-8 weeks which did not add significant prognostic value in their model. The prognostic value of CTCs was consistent across all subtypes of disease. Finally, in the multivariate analysis, CTC count was the strongest prognosticator for PFS and OS. Furthermore, continued elevation of CTCs is known to be associated with a poor prognosis as well as a sign of chemo-resistance. The prospective SWOGS0500 was designed to test the hypothesis that an early change in chemotherapy regimen in patients with positive CTCs (defined as ≥ 5 CTCs) after first cycle chemotherapy administered for previously untreated disease. A recent trial of MBC patients used persistently elevated CTCs as a marker of chemo-resistance and evaluated an early intervention of changing chemotherapy after one cycle to see if this would improve survival [1]. From October 1, 2006 until March 15, 2012, a total of 595 eligible and evaluable patients were enrolled in the study and started on first line therapy. Approximately 330 patients were classified as HR+/Her-2 negative (Luminal subtype) and 134 patients had triple negative breast cancer (TNBC). All patients received first-line chemotherapy. At baseline 276 patients (46%) had positive CTCs and 319 (54%) did not (arm A). Patients with HR+ disease had higher percentage of CTC positivity compared to TNBC (64 % vs. 21 % respectively). Of those patients with initially increased CTCs, 31 (10%) were not retested, 165 were evaluated at 3-4 weeks after chemotherapy and found to have normal (< 5 CTCs) and assigned to continuation of treatment (arm B), and the 123 patients with persistent CTC elevation were randomly assigned to either maintain same therapy (arm C1) or change therapy (arm C2). Interestingly, at the second time point the percentage of CTCs ≥ 5 in HR+ disease increased (72% vs. 22%). No difference in median OS was observed between arm C1 and C2 (10.7 and 12.5 months, respectively; $P = .98$). CTCs were strongly prognostic. Median OS for arms A, B, and C (C1 and C2 combined) were 35 months, 23 months, and 13 months, respectively ($P < .001$). The authors concluded that, those patients with continually elevated CTCs, irrespective of their subtype status, obtained limited therapeutic benefit from continuation or change of chemotherapy regimen. These patients had a shortened OS, and there was no

survival advantage to changing chemotherapy early versus continuing on initial therapy. The study again confirmed the prognostic value of CTCs: the group with no CTCs at baseline had the best median OS, and the group whose CTCs fell after 21 days of initiating chemotherapy fared better than the group with persistently elevated CTCs. This may also suggest that the negative prognostic impact of high CTC counts is not merely expression of the overt tumor burden, but may also reflect the presence of undetectable micrometastatic disease, and thus predict a greater tendency to the metastatic spread.

The prognostic and predictive value of CTCs in different subtypes has been investigated in multiple studies. A retrospective analysis by Munzone *et al* analyzed CTC count among the different molecular subtypes [11]. Again, they found that CTCs were predictive for OS for all subtypes. Generally the group of patients with 0 CTCs had better outcomes than those with 1-4 CTCs or >5 CTCs except this difference was not noted in the HER2+ group. Subsequently, two large prospective trials, with over 200 MBC patients each, found CTCs to be prognostic regardless of molecular subtype or prior therapy. Moreover, Giuliano *et al* reported that baseline detection of CTCs was associated with endocrine resistance [12]. These observations raise questions regarding the additional, prognostic information provided by CTCs that might not be limited to the number of CTCs only. For example, CTCs might reflect the primary tumor's biology, including intratumoral heterogeneity. Breast tumors are considered being ER α positive if 1% of the cells show nuclear reactivity of any intensity by immunohistochemical investigation [13, 14]. Therefore, CTCs arising from primary ER α -positive breast tumors are not necessarily expected to be ER α -positive. ER-negative CTCs might originate from ER-negative cells of ER-mosaic primary tumor [15], or ER-negative clones might be selected and get growth superiority under the pressure of anti-ER therapy [16]. Since endocrine treatment is dependent on the hormone receptor status and targets ER α -positive cancer cells only, CTC heterogeneity might be one reason for treatment failure and metastasis development in patients with ER α -positive tumors. Several investigators demonstrated a low percentage of ER+ CTCs in patients with HR+ MBC likely contributing to the limited benefit of endocrine therapy [17, 18]. Most recently we showed that HR+ MBC patients have a high incidence of heterogeneous expression of ER and cells with ER+ disease can also carry an *ESR1* mutation as expression of endocrine resistance (Figure 1).

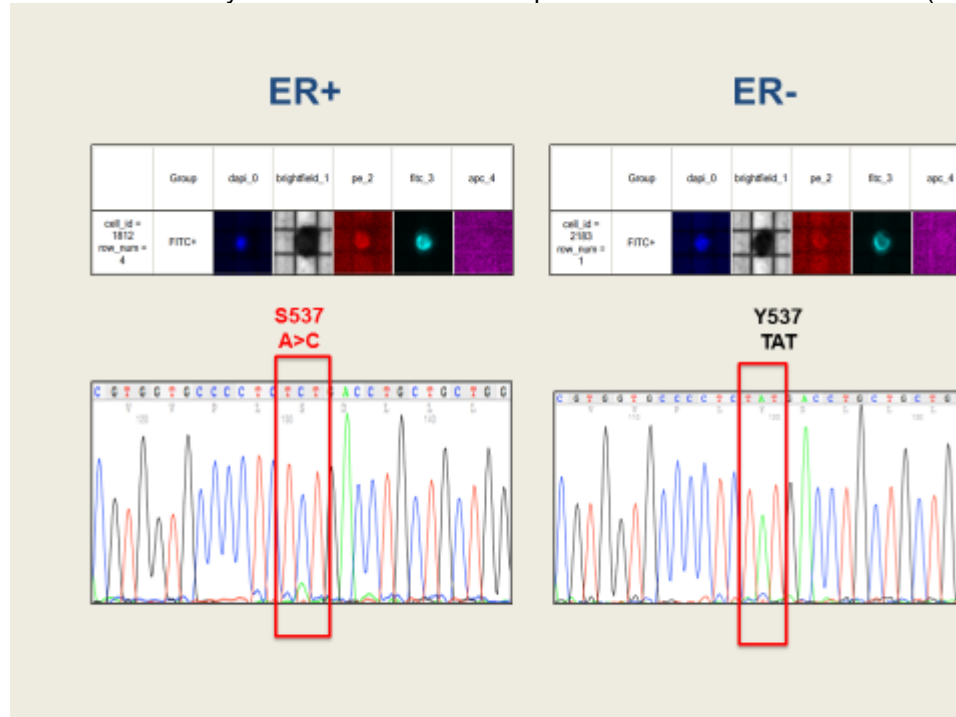


Figure 1.

In conclusion, new therapeutic modalities are needed for those patients with aggressive MBC as identified by CTC ≥ 5 irrespective of disease subtype and recurrent vs. *de-novo* status. Moreover, we recently showed that patients with elevated CTC counts have greater tendency for an early progression and particularly with development of new sites and visceral metastases [12] suggesting that treatments with the ability to reduce or eliminate CTCs can have a significant impact on outcome as measured by PFS and OS.

In summary we can conclude that:

- 1) Elevated baseline CTCs is an indication of aggressive disease with propensity to further progress with new lesions irrespective of disease subtype;
- 2) Hormone-receptor positive MBC with higher CTC detection likely reflects a disease resistant or refractory to endocrine therapy and with heterogeneous ER expression, possibly carrying *ESR1* mutations;
- 3) TNBC either recurrent or *de-novo* with higher CTCs identify a very aggressive cohort with intrinsic chemotherapy resistance.
- 4) There is a small number of patients with ≥ 5 CTCs that derives limited and temporary benefit from chemotherapy;
- 5) Chemo-refractory disease is associated with a very short overall survival and they need novel therapeutic approach.

In summary, we hypothesize that the use of chemotherapy combined with anti-PD-1 immunotherapy will reduce immune suppression and facilitate T-cell directed elimination of CTCs. The immune targeting of those cells will translate in therapeutic benefit as measured by improved PFS, overall response rate (ORR), clinical benefit rate (CBR) and ultimately OS. The CTCs should demonstrate a decline and a measurable objective response deriving also by the combination with cytotoxic chemotherapy.

1.2 Pharmaceutical and Therapeutic Background

The importance of intact immune surveillance in controlling outgrowth of neoplastic transformation has been known for decades. Accumulating evidence shows a correlation between tumor-infiltrating lymphocytes (TILs) in cancer tissue and favorable prognosis in various malignancies. In particular, the presence of CD8⁺ T-cells and the ratio of CD8⁺ effector T-cells / FoxP3⁺ regulatory T-cells seems to correlate with improved prognosis and long-term survival in many solid tumors.

The PD-1 receptor-ligand interaction is a major pathway hijacked by tumors to suppress immune control. The normal function of PD-1, expressed on the cell surface of activated T-cells under healthy conditions, is to down-modulate unwanted or excessive immune responses, including autoimmune reactions. PD-1 (encoded by the gene *Pdcd1*) is an Ig superfamily member related to CD28 and CTLA-4 which has been shown to negatively regulate antigen receptor signaling upon engagement of its ligands (PD-L1 and/or PD-L2). The structure of murine PD-1 has been resolved. PD-1 and family members are type I transmembrane glycoproteins containing an Ig Variable-type (V-type) domain responsible for ligand binding and a cytoplasmic tail which is responsible for the binding of signaling molecules. The cytoplasmic tail of PD-1 contains 2 tyrosine-based signaling motifs, an immunoreceptor tyrosine-based inhibition motif (ITIM) and an immunoreceptor tyrosine-based switch motif (ITSM). Following T-cell stimulation, PD-1 recruits the tyrosine phosphatases SHP-1 and SHP-2 to the ITSM motif within its cytoplasmic tail, leading to the dephosphorylation of effector molecules such as CD3 ζ , PKC θ and ZAP70 which are involved in the CD3 T-cell signaling cascade. The mechanism by which PD-1 down modulates T-cell responses is similar to, but distinct from that of CTLA-4 as both molecules regulate an overlapping set of signaling proteins. PD-1 was shown to be expressed on activated lymphocytes

including peripheral CD4+ and CD8+ T-cells, B-cells, T regs and Natural Killer cells. Expression has also been shown during thymic development on CD4-CD8- (double negative) T-cells as well as subsets of macrophages and dendritic cells. The ligands for PD-1 (PD-L1 and PD-L2) are constitutively expressed or can be induced in a variety of cell types, including non-hematopoietic tissues as well as in various tumors. Both ligands are type I transmembrane receptors containing both IgV- and IgC-like domains in the extracellular region and contain short cytoplasmic regions with no known signaling motifs. Binding of either PD-1 ligand to PD-1 inhibits T-cell activation triggered through the T-cell receptor. PD-L1 is expressed at low levels on various non-hematopoietic tissues, most notably on vascular endothelium, whereas PD-L2 protein is only detectably expressed on antigen-presenting cells found in lymphoid tissue or chronic inflammatory environments. PD-L2 is thought to control immune T-cell activation in lymphoid organs, whereas PD-L1 serves to dampen unwarranted T-cell function in peripheral tissues. Although healthy organs express little (if any) PD-L1, a variety of cancers were demonstrated to express abundant levels of this T-cell inhibitor. PD-1 has been suggested to regulate tumor-specific T-cell expansion in subjects with melanoma. This suggests that the PD-1/PD-L1 pathway plays a critical role in tumor immune evasion and should be considered as an attractive target for therapeutic intervention.

Pembrolizumab is a potent and highly selective humanized monoclonal antibody (mAb) of the IgG4/kappa isotype designed to directly block the interaction between PD-1 and its ligands, PD-L1 and PD-L2. Pembrolizumab (Keytruda™) has recently been approved in the United States for the treatment of patients with unresectable or metastatic melanoma and disease progression following ipilimumab and, if BRAF V600 mutation positive, a BRAF inhibitor.

1.2.1 *Preclinical and Clinical Trial Data*

Refer to the Investigator's Brochure for Preclinical and detailed Clinical data. Below is a summary of the safety data.

Immune-Mediated Pneumonitis

Immune-mediated pneumonitis, including fatal cases, occurred in patients receiving pembrolizumab. Monitor patients for signs and symptoms of pneumonitis. Evaluate patients with suspected pneumonitis with radiographic imaging and administer corticosteroids (initial dose of 1 to 2 mg/kg/day prednisone or equivalent followed by a taper) for Grade 2 or greater pneumonitis. Withhold pembrolizumab for moderate (Grade 2) pneumonitis, and permanently discontinue pembrolizumab for severe (Grade 3), life-threatening (Grade 4), or recurrent moderate (Grade 2) pneumonitis. Melanoma Pneumonitis occurred in 32 (2.0%) of 1567 patients receiving pembrolizumab in Trials 1, 2, and 6, including Grade 1 (0.8%), Grade 2 (0.8%), and Grade 3 (0.4%) pneumonitis. The median time to development of pneumonitis was 4.3 months (range: 2 days to 19.3 months). The median duration was 2.6 months (range: 2 days to 15.1 months). Twelve (38%) of the 32 patients received corticosteroids, with 9 of the 12 receiving high-dose systemic corticosteroids for a median duration of 8 days (range: 1 day to 1.1 months) followed by a corticosteroid taper. Pneumonitis led to discontinuation of pembrolizumab in 9 (0.6%) patients. Pneumonitis completely resolved in 21 (66%) of the 32 patients. Non-small cell lung cancer (NSCLC) Pneumonitis occurred in 19 (3.5%) of 550 patients with NSCLC, including Grade 2 (1.1%), Grade 3 (1.3%), Grade 4 (0.4%), or Grade 5 (0.2%) pneumonitis in patients receiving pembrolizumab in Trial 1. The median time to development of pneumonitis was 1.7 months (range: 4 days to 12.9 months). In patients receiving pembrolizumab 10 mg/kg every 14 days, the median time to development of pneumonitis was shorter (1.5 months) compared with patients receiving 10 mg/kg every 21 days (3.5 months). Sixteen of the 19 patients (84%) received corticosteroids, with 14 of the 19 (74%) requiring high-dose systemic corticosteroids (greater than or equal to 40 mg prednisone or equivalent per day). The median starting dose of high-dose corticosteroid treatment for these fourteen patients was 60 mg/day with a median duration of treatment of 8 days (range: 1 day to 4.2 months). The median duration of pneumonitis was 1.2 months (range: 5 days to 12.4 months). Pneumonitis occurred more

frequently in patients with a history of asthma/chronic obstructive pulmonary disease (5.4%) than in patients without a history of these diseases (3.1%). Pneumonitis occurred more frequently in patients with a history of prior thoracic radiation (6.0%) than in patients who did not receive prior thoracic radiation (2.6%). Pneumonitis led to discontinuation of pembrolizumab in 12 (2.2%) patients. Pneumonitis completely resolved in 9 patients. Pneumonitis was reported as ongoing in 9 patients and one patient with ongoing pneumonitis died within 30 days of the last dose of pembrolizumab.

Immune-Mediated Colitis

Immune-mediated colitis occurred in patients receiving pembrolizumab. Monitor patients for signs and symptoms of colitis. Administer corticosteroids (initial dose of 1 to 2 mg/kg/day prednisone or equivalent followed by a taper) for Grade 2 or greater colitis. Withhold pembrolizumab for moderate (Grade 2) or severe (Grade 3) colitis, and permanently discontinue pembrolizumab for life-threatening (Grade 4) colitis. Melanoma Colitis occurred in 31 (2.0%) of 1567 patients receiving pembrolizumab in Trials 1, 2, and 6, including Grade 2 (0.5%), Grade 3 (1.1%), and Grade 4 (0.1%) colitis. The median time to onset of colitis was 3.4 months (range: 10 days to 9.7 months). The median duration of colitis was 1.4 months (range: 1 day to 7.2 months). Twenty-one (68%) of the 31 patients received corticosteroids, all of whom required high-dose systemic corticosteroids for a median duration of 6 days (range: 1 day to 5.3 months) followed by a 5 corticosteroid taper. Colitis led to discontinuation of pembrolizumab in 14 (0.9%) patients. Colitis resolved in 27 (87%) of the 31 patients. NSCLC Colitis occurred in 4 (0.7%) of 550 patients, including Grade 2 (0.2%) or Grade 3 (0.4%) colitis in patients receiving pembrolizumab in Trial 1. The median time to onset of colitis was 1.6 months (range: 28 days to 2.2 months) and the median duration was 16 days (range: 7 days to 1.3 months). Two patients were started on high-dose corticosteroids (greater than or equal to 40 mg prednisone or equivalent per day) and two patients were started on low dose corticosteroids. One patient (0.2%) discontinued pembrolizumab due to colitis. Three patients with colitis experienced complete resolution of the event.

Immune-Mediated Hepatitis

Immune-mediated hepatitis occurred in patients receiving pembrolizumab. Monitor patients for changes in liver function. Administer corticosteroids (initial dose of 0.5 to 1 mg/kg/day [for Grade 2 hepatitis] and 1 to 2 mg/kg/day [for Grade 3 or greater hepatitis] prednisone or equivalent followed by a taper) and, based on severity of liver enzyme elevations, withhold or discontinue pembrolizumab. Melanoma Hepatitis occurred in 16 (1.0%) of 1567 patients receiving pembrolizumab in Trials 1, 2, and 6, including Grade 2 (0.1%), Grade 3 (0.7%), and Grade 4 (0.1%) hepatitis. The time to onset was 26 days (range: 8 days to 21.4 months). The median duration was 1.2 months (range: 8 days to 4.7 months). Eleven (69%) of the 16 patients received corticosteroids, with 10 of the 11 receiving high-dose systemic corticosteroids for a median duration of 5 days (range: 1 to 14 days) followed by a corticosteroid taper. Hepatitis led to discontinuation of pembrolizumab in 6 (0.4%) patients. Hepatitis resolved in 14 (88%) of the 16 patients.

Immune-Mediated Endocrinopathies

Hypophysitis occurred in patients receiving pembrolizumab. Monitor for signs and symptoms of hypophysitis (including hypopituitarism and adrenal insufficiency). Administer corticosteroids and hormone replacement as clinically indicated. Withhold pembrolizumab for moderate (Grade 2) hypophysitis and withhold or discontinue pembrolizumab for severe (Grade 3) or life-threatening (Grade 4) hypophysitis. Melanoma Hypophysitis occurred in 13 (0.8%) of 1567 patients receiving pembrolizumab in Trials 1, 2, and 6 including Grade 2 (0.3%), Grade 3 (0.3%), and Grade 4 (0.1%) hypophysitis. The time to onset was 3.3 months (range: 1 day to 7.2 months). The median duration was 2.7 months (range: 12 days to 12.7 months). Twelve (92%) of the 13 patients received corticosteroids, with 4 of the 12 patients receiving high-dose systemic corticosteroids. Hypophysitis led to discontinuation of pembrolizumab in 4

(0.3%) patients. Hypophysitis resolved in 7 (54%) of the 13 patients. In Trial 1, NSCLC hypophysitis occurred in 1 (0.2%) of 550 patients, which was Grade 3 in severity. The time to onset was 3.7 months. The patient was treated with systemic corticosteroids and physiologic hormone replacement therapy. The patient did not discontinue pembrolizumab due to hypophysitis. Thyroid Disorders Thyroid disorders can occur at any time during treatment. Monitor patients for changes in thyroid function (at the start of treatment, periodically during treatment, and as indicated based on clinical evaluation) and for clinical signs and symptoms of thyroid disorders. Administer replacement hormones for hypothyroidism and manage hyperthyroidism with thionamides and beta-blockers as appropriate. Withhold or discontinue pembrolizumab for severe (Grade 3) or life-threatening (Grade 4) hyperthyroidism. Melanoma Hyperthyroidism occurred in 51 (3.3%) of 1567 patients receiving pembrolizumab in Trials 1, 2, or 6, including Grade 2 (0.6%) and Grade 3 (0.1%) hyperthyroidism. The median time to onset was 1.4 months (range: 1 day to 21.9 months). The median duration was 1.7 months (range: 1 day to 12.8 months). Hyperthyroidism led to discontinuation of pembrolizumab in 2 (0.1%) patients. Hyperthyroidism resolved in 36 (71%) of the 51 patients. Hypothyroidism occurred in 127 (8.1%) of 1567 patients receiving pembrolizumab in Trials 1, 2, and 6 including Grade 3 (0.1%) hypothyroidism. The median time to onset of hypothyroidism was 3.3 months (range: 5 days to 18.9 months). The median duration was 5.4 months (range: 6 days to 24.3 months). No patients discontinued pembrolizumab due to hypothyroidism. Hypothyroidism resolved in 24 (19%) of the 127 patients. NSCLC Hyperthyroidism occurred in 10 (1.8%) of 550 patients receiving pembrolizumab in Trial 1, including Grade 2 (0.7%) or Grade 3 (0.3%) hyperthyroidism. The median time to onset was 1.8 months (range: 2 days to 3.4 months), and the median duration was 4.5 months (range: 4 weeks to 7.5 months). No patients discontinued pembrolizumab due to hyperthyroidism. Hypothyroidism occurred in 38 (6.9%) of 550 patients receiving pembrolizumab in Trial 1, including Grade 2 (5.5%) or Grade 3 (0.2%) hypothyroidism. The median time to onset was 4.2 months (range: 20 days to 11.2 months), and the median duration was 5.8 months (range: 11 days to 22.8 months). No patients discontinued pembrolizumab due to hypothyroidism. Type 1 diabetes mellitus, including diabetic ketoacidosis, occurred in 3 (0.1%) of 2117 patients with melanoma or NSCLC receiving pembrolizumab in Trials 1, 2, and 6. Monitor patients for hyperglycemia or other signs and symptoms of diabetes. Administer insulin for type 1 diabetes, and withhold pembrolizumab and administer anti-hyperglycemics in patients with severe hyperglycemia.

Immune-Mediated Nephritis and Renal Dysfunction

Immune-mediated nephritis occurred in patients receiving pembrolizumab. Monitor patients for changes in renal function. Administer corticosteroids (initial dose of 1 to 2 mg/kg/day prednisone or equivalent followed by a taper) for Grade 2 or greater nephritis. Withhold pembrolizumab for moderate (Grade 2), and permanently discontinue pembrolizumab for severe (Grade 3) or life-threatening (Grade 4) nephritis. Melanoma Nephritis occurred in 7 (0.4%) of 1567 patients receiving pembrolizumab in Trials 1, 2, and 6, including Grade 2 (0.2%), Grade 3 (0.2%), and Grade 4 (0.1%) nephritis. The median time to onset of nephritis was 5.1 months (range: 12 days to 12.8 months). The median duration was 1.1 months (range: 3 days to 3.3 months). Six (86%) of the 7 patients received corticosteroids, with 5 of the 6 receiving high-dose systemic corticosteroids for a median duration of 15 days (range: 3 days to 1.6 months) followed by a corticosteroid taper. Nephritis led to discontinuation of pembrolizumab in 2 (0.1%) patients. Nephritis resolved in 4 (57%) of the 7 patients.

Other Immune-Mediated Adverse Reactions

Other clinically important immune-mediated adverse reactions can occur. 7 For suspected immune-mediated adverse reactions, ensure adequate evaluation to confirm etiology or exclude other causes. Based on the severity of the adverse reaction, withhold pembrolizumab and administer corticosteroids. Upon improvement to Grade 1 or less,

initiate corticosteroid taper and continue to taper over at least 1 month. Based on limited data from clinical studies in patients whose immune-related adverse reactions could not be controlled with corticosteroid use, administration of other systemic immunosuppressants can be considered. Resume pembrolizumab when the immune-mediated adverse reaction remains at Grade 1 or less following corticosteroid taper. Permanently discontinue pembrolizumab for any Grade 3 immune-mediated adverse reaction that recurs and for any life-threatening immune-mediated adverse reaction. The following clinically significant Melanoma immune-mediated adverse reactions occurred in less than 1% (unless otherwise indicated) of 1567 patients with melanoma treated with pembrolizumab in Trials 1, 2, and 6: arthritis (1.6%), exfoliative dermatitis, bullous pemphigoid, uveitis, myositis, Guillain-Barré syndrome, myasthenia gravis, vasculitis, pancreatitis, hemolytic anemia, and partial seizures arising in a patient with inflammatory foci in brain parenchyma. The following clinically significant, immune-mediated adverse reactions occurred in less than 1% of 550 patients with NSCLC treated with pembrolizumab in Trial 1: rash, vasculitis, hemolytic anemia, serum sickness, and myasthenia gravis.

1.2.2 *Rationale for Dose Selection/Regimen/Modification*

Pembrolizumab

An open-label Phase I trial (Protocol 001) is being conducted to evaluate the safety and clinical activity of single agent pembrolizumab. The dose escalation portion of this trial evaluated three dose levels, 1 mg/kg, 3 mg/kg, and 10 mg/kg, administered every 2 weeks (Q2W) in subjects with advanced solid tumors. All three dose levels were well tolerated and no dose-limiting toxicities were observed. This first in human study of pembrolizumab showed evidence of target engagement and objective evidence of tumor size reduction at all dose levels (1 mg/kg, 3 mg/kg and 10 mg/kg Q2W). No maximum tolerated dose (MTD) has been identified to date. Recent data from other clinical studies within the pembrolizumab program has shown that a lower dose of pembrolizumab and a less frequent schedule may be sufficient for target engagement and clinical activity.

Pharmacokinetic (PK) data analysis of pembrolizumab administered Q2W and Q3W showed slow systemic clearance, limited volume of distribution, and a long half-life (refer to Investigator Brochure). Pharmacodynamic data (IL-2 release assay) suggested that peripheral target engagement is durable (>21 days). This early PK and pharmacodynamic data provides scientific rationale for testing a Q2W and Q3W dosing schedule.

A population PK analysis has been performed using serum concentration time data from 476 patients. Within the resulting population PK model, clearance and volume parameters of pembrolizumab were found to be dependent on body weight. The relationship between clearance and body weight, with an allometric exponent of 0.59, is within the range observed for other antibodies and would support both body weight normalized dosing or a fixed dose across all body weights. Pembrolizumab has been found to have a wide therapeutic range based on the melanoma indication. The differences in exposure for a 200 mg fixed dose regimen relative to a 2 mg/kg Q3W body weight based regimen are anticipated to remain well within the established exposure margins of 0.5 – 5.0 for pembrolizumab in the melanoma indication. The exposure margins are based on the notion of similar efficacy and safety in melanoma at 10 mg/kg Q3W vs. the proposed dose regimen of 2 mg/kg Q3W (i.e. 5-fold higher dose and exposure). The population PK evaluation revealed that there was no significant impact of tumor burden on exposure. In addition, exposure was similar between the NSCLC and melanoma indications. Therefore, there are no anticipated changes in exposure between different indication settings.

The rationale for further exploration of 2 mg/kg and comparable doses of pembrolizumab in solid tumors is based on: 1) similar efficacy and safety of pembrolizumab when dosed

at either 2 mg/kg or 10 mg/kg Q3W in melanoma patients, 2) the flat exposure-response relationships of pembrolizumab for both efficacy and safety in the dose ranges of 2 mg/kg Q3W to 10 mg/kg Q3W, 3) the lack of effect of tumor burden or indication on distribution behavior of pembrolizumab (as assessed by the population PK model) and 4) the assumption that the dynamics of pembrolizumab target engagement will not vary meaningfully with tumor type.

The choice of the 200 mg Q3W as an appropriate dose for the switch to fixed dosing is based on simulations performed using the population PK model of pembrolizumab showing that the fixed dose of 200 mg every 3 weeks will provide exposures that 1) are optimally consistent with those obtained with the 2 mg/kg dose every 3 weeks, 2) will maintain individual patient exposures in the exposure range established in melanoma as associated with maximal efficacy response and 3) will maintain individual patients exposure in the exposure range established in melanoma that are well tolerated and safe.

A fixed dose regimen will simplify the process and be more convenient for physicians, which will reduce the potential for dosing errors. A fixed dosing scheme will also reduce complexity in the logistical chain at treatment facilities and reduce wastage.

Carboplatin

Patients included in the study are TNBC with CTCs+. Those patients have been exposed to taxanes in the adjuvant or neoadjuvant setting. Furthermore, the prospective SWOG study evaluating the impact of chemotherapy change in patients with detectable CTCs at time of recurrence and known for the negative impact of this strategy confirmed previous observations [1]. Patients with CTCs do not benefit from single agent taxanes as first-line strategy and therefore, other agents should be selected for these patients. The combination of NGS and CTC detection and NGS suggests a correlation with higher mutational rate and particular association with *TP53* and cell cycle associated genes. Platinum salts have demonstrated activity in MBC. A retrospective study by Sirohi et al. reviewed 155 patients with metastatic breast cancer (34 with TNBC) treated with platinum-based chemotherapy [19]. A little gain in progression-free survival (PFS) (6 vs. 4 months, $p = 0.05$), but no significant differences in OS and overall response rate (41% vs. 31%, $p = 0.3$) were observed in TNBC patients compared to non-TNBC. The TBCRC009 study demonstrated single agent activity of both cisplatin and carboplatin in first-line TNBC [20]. The TNT trial is the first randomized phase III trial testing the activity of single-agent platinum chemotherapy against a standard chemotherapy with single-agent taxane for TNBC patients or BRCA1/2 mutation carriers [21]. Three hundred seventy-six patients were enrolled in the study and randomly assigned to receive carboplatin or docetaxel for 6–8 cycles, with crossover possibility on progression. After a median follow-up of 11 months, there was no significant improvement in overall response rate for carboplatin over docetaxel (31.4% vs. 35.6%, respectively; $p = 0.44$). However, among patients carrying a germline BRCA1/2 mutation ($n = 48$), the objective response rate was more than two-fold higher when treated with carboplatin (68% vs. 33.3%, $p = 0.03$), with a longer PFS (6.8 vs. 3.1 months). In contrast, docetaxel had a median PFS of 4.7 months, with no differences according to BRCA1/2 mutational status.

1.2.3 Rationale for Efficacy Endpoints

Elevated baseline CTCs in recurrent or *de-novo* MBC is an indication of aggressive disease with propensity to further progress with new lesions irrespective of disease subtype. Previous studies demonstrated that MBC with ≥ 5 CTCs is primarily reflecting chemo-resistant disease associated with a very short PFS and OS and those patients are in need of novel therapeutic approaches. Furthermore, we demonstrated that MBC CTC detection is associated with higher levels of specific genomic abnormalities in metastatic tissue analyzed with NGS. Because of the high expression of PD-L1 in CTCs isolated

from patients with MBC, particularly TNBC we hypothesize that immune targeting of PD-1 would restore activity of CD8 and increase antitumor effects of chemotherapy resulting in improved efficacy, primarily affecting OS. The lack of novel therapeutics in TNBC (compared to HR+ MBC) and the particularly poor-outcome in TNBC CTCs+ support the need to early implementation of checkpoint inhibitors in this subtype. Moreover, CTCs express markers of immune suppression (e.g. PD-L1).

In addition, tumor response will be assessed by the irRECIST for investigational purposes, not assessment of primary objective [22]. Tumor response by irRECIST is defined as an immune-related PR (irPR) or CR (irCR) over a period of at least 4 weeks. We will define immune-related clinical benefit rate as immune-related stable disease (irSD), irPR, or irCR.

1.3 Rationale for Current Study

MBC is a heterogeneous disease and long-term patient outcome can be influenced by various biological features, as well as by the extent and site of metastatic disease. Various molecular markers and blood-based tests have been investigated as surrogate for more aggressive disease, among them the most reliable appears enumeration of CTCs [4, 5]. Elevated baseline CTCs in MBC is an indication of aggressive disease with propensity to further progress with new lesions irrespective of disease subtype. Moreover, we recently showed that patients with elevated CTC counts have greater tendency for an early progression and particularly with development of new sites and visceral metastases [12] suggesting that treatments with the ability to reduce or eliminate CTCs can have a significant impact on outcome. CTCs express markers of immune suppression (e.g. PD-L1) and blockade of the programmed cell death 1 (PD-1) and its ligand (PD-L1) with pembrolizumab may result in improve disease control and outcome as measured by PFS and OS.

1.4 Rationale for Exploratory Studies

1.4.1 *The association of CTC detection with specific genomic abnormalities in metastatic tissue analyzed by next-genomic sequencing (NGS).*

We reasoned that this approach will identify potential therapeutic targets and/or the level of genomic instability associated with CTC-positive disease. Studies have shown that tumors with increased number of mutations tend to have increased T-cell dysfunction or immune suppression and may benefit from immune therapy aimed at restoring T-cell function.

In fact, cancer cells within the same tumor can exhibit different genotypes and the development of NGS methods helped to provide fundamental insight on the genetic intra-tumor heterogeneity [23]. This clonal heterogeneity confers greater resistance to selective environmental pressures and, most of all, to systemic treatments [24]. Therefore, based on these preliminary considerations we decided to analyze the correlation between CTC enumeration and genomic alterations in tumor tissue of MBC patients.

We performed a preliminary analysis of patients with MBC that were enrolled in a prospective clinical trial evaluating blood-based biomarkers including CTCs. Prospective enumeration of CTCs was performed using the CellSearch® system (Janssen Diagnostics). Recurrent lesions were biopsied and sent for NGS analysis by FoundationOne® testing (Foundation Medicine, Cambridge, MA), which sequences 236 genes known to be somatically altered in solid tumors. Patients with both tests were selected for this analysis. CTCs were measured in 8 peripheral blood samples drawn from 7 MBC patients. HER2-positive tumors were excluded because of the lack of prediction of CTC count in disease treated with HER2-targeted therapy. Eight tissue samples from seven patients were sent for NGS which revealed 38 unique mutations with a total of 53 mutational events. The most common mutations/alterations were TP53

(86%), MYC amplification (57%), PIK3CA (29%) and PTEN loss (29%). Of the seven patients, five had CTC<5 (0, 0, 1, 2, 3) whereas three patients had CTC>5 (10, 11, 22). The range of CTCs detected was 0 to 22 with an average of 6. For the patients who had less than 5 CTC detected, the respective tissues showed ≤ 2 gene mutations or losses. The other 3 samples, taken from 2 different patients who had more than 5 CTCs, had > 3 gene mutations or losses detected on tumor tissues analysis. Additional genomic abnormality (e.g. amplifications) distribution was evaluated. The data showed an association between specific molecular abnormalities and CTC enumeration, specifically we found that all patients with ≥ 5 CTCs demonstrated PTEN loss in tissue specimens, while all patients with < 5 CTCs expressed TP53 mutations. Moreover, 3 out of 5 patients with < 5 CTCs showed MYC amplification. The patient who had two blood samples sent for CTC analysis had persistently elevated CTCs (10, 11) which correlated with her progression of disease prompting a change in therapy.

1.4.2 *The association of CTC detection with activation of innate immunity and T-cell dysfunction.*

Previous studies have demonstrated abnormal immune functions in patients with advanced solid tumors including breast cancer [25, 26]. In particular, CD4+CD25+FOXP3+ and CD8+CD28– Tregs are usually elevated in cancer patients. Chemotherapy was traditionally thought to be deleterious to anti-tumor immunity, but some chemotherapy agents may have the potential to augment anti-cancer immunity by inhibiting immune tolerance and suppression [27-30]. Paclitaxel has been found to inhibit the number and function of CD4+CD25+FOXP3+ Tregs directly [27]. In addition, cyclophosphamide was reported to inhibit the activity of CD4+CD25+ Tregs [8], and the generation and function of CD8+CD28– Tregs [28, 29]. In contrast, arsenic trioxide, dacarbazine, 5-fluorouracil and methotrexate increase the frequency of Tregs in peripheral blood mononuclear cells activated with T-cell mitogen [30]. The combination of gemcitabine and cisplatin demonstrated the ability to reduce CD4+CD25+FOXP3+ and CD8+CD28– Tregs at 3 weeks after administration of first dose [31]. The above data suggest that chemotherapeutic agents may have varying effects on Treg populations and activity. It has been postulated that combining some of these chemotherapeutic agents with immune-checkpoint regulators could decrease immune suppression and increase the efficacy of immune therapy [32]. Theoretically, the concomitant administration of chemotherapy and immune-therapy may not necessarily represent the optimal solution for achieving maximum benefit and alternative solutions should be explored to maximize synergistic effect. More recently it has been proposed that in patients with recurrent disease and reduced likelihood of response to chemotherapy an initial treatment with immune-therapy can “prime” the cancer cells to a more effective cytotoxic effect of chemotherapy. Therefore, patients with detectable CTCs would likely benefit from a “window” treatment with immune therapy before receiving combination regimen [33].

Our group has been investigating the possibility of identifying members of the innate immune system in the peripheral blood of patients with solid tumors to understand the complex relationship with cancer cells. We performed a prospective study using two different methods of selection for the identification of CTCs, the FDA-approved CellSearch® and a microfiltration system (CellSieve™). During this study we identified cancer-associated macrophage-like cells (CAMLs) that appear to indicate a more aggressive disease in metastatic conditions [34]. More recently, we demonstrated the expression of immune checkpoint pathways, including the programmed cell death protein-1 (PD-1) and the ligand B7-H1/PD-L1 [Figure 2].

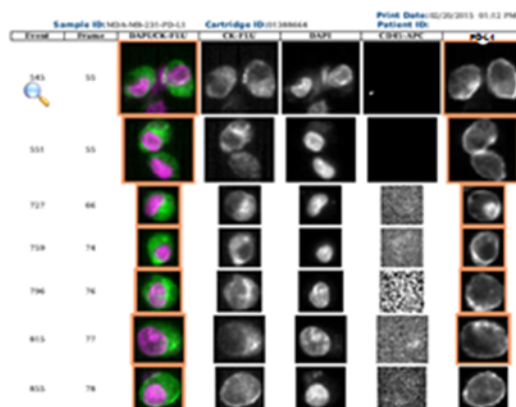


Figure 2.

These initial data demonstrate a predominant expression of PD-L1 in CTCs indicating a role in immune suppression and the potential for therapeutic targeting.

In conclusion, our preliminary analysis suggests that tumors with greater genetic instability, as demonstrated by higher number of mutations on NGS, and greater metastatic potential are associated with higher numbers of CTCs in the peripheral blood. Moreover, CTCs detection is associated with CAML and expression of PDL-1 suggesting participation of immune activation in the metastatic process. Taken together the evidence indicates that CTC detection can be used to select patients with greater chance of immune dysfunction and that would likely benefit from immune therapies.

2 OBJECTIVES

2.1 Primary Objective(s)

Evaluate the impact on PFS of the combination pembrolizumab - carboplatin in patients with CTC positive triple negative (TN) MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV. PFS will be measured at 9 weeks after beginning treatment.

Hypothesis:

- Elevated baseline CTCs in MBC is an indication of aggressive disease with propensity to further progress with new lesions irrespective of disease subtype;
- TNBC with CTCs detection is the most aggressive and lethal form of MBC with associated CTCs-PDL-1+.
- CTC detection, particularly in TNBC is associated with higher levels of genomic abnormalities in metastatic tissue analyzed by NGS;
- CTCs express markers of immune suppression (e.g. PD-L1);

- Blockade of the programmed cell death 1 (PD-1) and its ligand (PD-L1) with pembrolizumab should result in improved disease control and outcome as measured by PFS and OS.

2.2 Secondary Objective(s)

- 2.2.1 *To determine the toxicity and safety of the combination pembrolizumab - carboplatin in patients with CTC positive, triple negative (TN) MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV. Adverse Events will be assessed according to the NCI CTCAE 4.03.*
- 2.2.2 *Evaluate the impact on OS of the combination Carboplatin - pembrolizumab in patients with CTC positive TN, MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV.*
- 2.2.3 *To assess the ORR and CBR according to RECIST criteria in patients with Carboplatin - pembrolizumab in patients with CTC positive TN MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV.*
- 2.2.4 *To assess immune-related response using tumor response by irRECIST as irPR or irCR. We will define immune-related clinical benefit rate as immune-related stable disease (irSD), irPR, or irCR.*
- 2.2.5 *Measure the time to new metastases (TTNM)*
- 2.2.6 *Evaluate ORR and clinical benefit in relation to PDL-1 expression in tissue and CTC diversity in antigen-specific T-cell receptors (TCR)*

Hypothesis:

- The detection of CTCs in TN MBC is associated with treatment-resistant disease and development of new metastatic sites. The ability to interfere with the metastatic process can result in targeting of CTCs affecting the development of new metastases ultimately contributing to improved survival.

2.3 Exploratory Objective

To be measured at baseline, cycles 3 and 4 and at the end of treatment.

- 2.3.1 *Measure immune biomarkers (PDL-1) in CTCs (CellSearch™) and immune cells such as CAMLs (CellSieve™) and correlate with therapeutic benefit;*

3 PATIENT ELIGIBILITY

Target population for this study are patients with CTC positive, triple negative (TN) MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV.

This will be a single-center trial at the Northwestern University Robert H.Lurie Comprehensive cancer Center at Chicago.. Patients will be recruited by the clinical team at Lynn Sage Comprehensive Breast Center of Northwestern University. The investigators may also utilize the services of the Cancer Trial Recruitment Nurse Coordinator as needed to assist with recruitment.

A total of 38 subjects (32 evaluable) will be needed for this trial. Approximately 8-10 potentially eligible patients are seen per month, and it is anticipated that at least 2 per month will be

accrued. Potential patients may be referred to the Principal Investigator (PI) at Northwestern University, Dr. Lisa E. Flaum, at 312-696-0990
Eligibility will be evaluated by the study team according to the following criteria. Eligibility waivers are not permitted. Subjects must meet all of the inclusion and none of the exclusion criteria to be registered to the study. Study treatment may not begin until a subject is registered. Please refer to Section 10 for complete instructions regarding registration procedures.

3.1 Subject Inclusion Criteria

- 3.1.1 *Patients must be **Hormone receptor (HR) negative and HER-2 negative (TNBC) metastatic breast cancer** and have not received prior chemotherapy for metastatic disease or be de-novo stage IV. Demonstrated HER-2 negative MBC (0 or 1+ by IHC or non-amplified by FISH) according to ASCO/CAPA guidelines.*
- 3.1.2 Patients should be either Stage IV de-novo or develop recurrence after completion of neoadjuvant/adjuvant therapy
- 3.1.3 Patients must be CTC positive (defined as CTCs ≥ 5)
- 3.1.4 Be ≥ 18 years of age on day of signing informed consent.
- 3.1.5 Have measurable disease based on RECIST 1.1 (see Appendix 12.3).
- 3.1.6 Be willing to provide archival tissue (if available) for correlative studies.
Note: The archived tumor tissue specimens may be from metastatic tumor specimen (first choice). In alternative, we can consider tissue from prior surgery or from prior diagnostic biopsy (second choice). Unavailability of archived tissue will not render subject ineligible for study.
- 3.1.7 Have a performance status of 0 or 1 on the ECOG Performance Status (see Appendix 12.1).
- 3.1.8 Demonstrate adequate organ function as defined in the table below, all screening labs should be performed within 14 days prior to registration.

Table 1.

System	Laboratory Value
Hematological	
Absolute neutrophil count (ANC)	$\geq 1,500$ /mcL
Platelets	$\geq 100,000$ / mcL
Hemoglobin	≥ 9 g/dL or ≥ 5.6 mmol/L without transfusion or EPO dependency (within 7 days of assessment)
Renal	
Serum creatinine OR Measured or calculated ^a creatinine clearance (GFR can also be used in place of creatinine or CrCl)	≤ 1.5 X upper limit of normal (ULN) OR ≥ 60 mL/min for subject with creatinine levels > 1.5 X institutional ULN
Hepatic	
Serum total bilirubin	≤ 1.5 X ULN OR
	Direct bilirubin $\leq$ ULN for subjects with total bilirubin levels > 1.5 ULN
AST (SGOT) and ALT (SGPT)	≤ 2.5 X ULN
Albumin	≥ 2.5 mg/dL
Coagulation	

International Normalized Ratio (INR) or Prothrombin Time (PT)	≤1.5 X ULN unless subject is receiving anticoagulant therapy as long as PT or PTT is within therapeutic range of intended use of anticoagulants
Activated Partial Thromboplastin Time (aPTT)	≤1.5 X ULN unless subject is receiving anticoagulant therapy as long as PT or PTT is within therapeutic range of intended use of anticoagulants
^a Creatinine clearance should be calculated per institutional standard.	

- 3.1.9 For patients with a known history of chronic hepatitis B virus (HBV) infection, the HBV viral load must be undetectable on suppressive therapy, if indicated.
- 3.1.10 Patients with a known history of hepatitis C virus (HCV) infection must have been treated and cured. For patients with HCV infection who are currently on treatment, they are eligible if they have an undetectable HCV viral load.
- 3.1.11 Female subject of childbearing potential (FOCBP) should have a negative urine or serum pregnancy within 7 days prior to registration; and **must be repeated within 3 days (72 hours) prior to first dose of study drug**. If the urine test is positive or cannot be confirmed as negative, a serum pregnancy test will be required.
(Note: A FOCBP is any woman (regardless of sexual orientation, having undergone a tubal ligation, or remaining celibate by choice) who meets the following criteria:
- Has not undergone a hysterectomy or bilateral oophorectomy
 - Has had menses at any time in the preceding 12 consecutive months (and therefore has not been naturally postmenopausal for > 12 months)
- 3.1.12 Female subjects of childbearing potential must be willing to use an adequate method of contraception as outlined in Section 4.5. Contraception must be used for the course of the study through 120 days after the last dose of study medication.
Note: Abstinence is acceptable if this is the usual lifestyle and preferred contraception for the subject.
- 3.1.13 Male subjects should agree to use an adequate method of contraception outlined in Section 4.5 starting with the first dose of study therapy through 120 days after the last dose of study therapy.
Note: Abstinence is acceptable if this is the usual lifestyle and preferred contraception for the subject.
- 3.1.14 Be willing to comply with all study procedures and able to provide written informed consent/assent for the trial.

3.2 Subject Exclusion Criteria

- 3.2.1 Histologically or cytologically confirmed HER2-positive (3+ by IHC or amplified by FISH) according to ASCO/CAP guidelines
Note: A HER2 result of 2+ by ICH is equivocal and requires a reflex test (same specimen using the alternative test) or new test (new specimen, if available, using same or alternative test)

- 3.2.2 Is currently participating and receiving study therapy or has participated in a study of an investigational agent and received study therapy or used an investigational device $\leq$ 28 days of registration.
- 3.2.3 Patients who had prior carboplatin in the metastatic setting are not eligible.
Note: Patients who have been exposed to carboplatin in neoadjuvant or adjuvant setting will be allowed to enroll, if they have progressed $\geq$ 6 months from completion of treatment
- 3.2.4 Has a diagnosis of immunodeficiency or is receiving systemic steroid therapy or any other form of immunosuppressive therapy $\leq$ 7 days prior to registration
Note: steroids as premedication are permitted
- 3.2.5 Has a known history of active TB (Bacillus Tuberculosis).
- 3.2.6 Hypersensitivity to pembrolizumab or any of its excipients.
- 3.2.7 Has had a prior anti-cancer monoclonal antibody (mAb) within 4 weeks prior to study Day 1 or who has not recovered (i.e., $\leq$ Grade 1 or baseline) from adverse events (AEs) due to agents administered more than 28 days earlier.
- 3.2.8 Has had prior chemotherapy, targeted small molecule therapy, or radiation therapy within 14 days prior to registration or who has not recovered (i.e., $\leq$ Grade 1 or at baseline) from AEs due to a previously administered agent.
Note: Subjects with $\leq$ Grade 2 neuropathy are an exception to this criterion and may qualify for the study.
- 3.2.9 If subject received major surgery, they must have recovered adequately from the toxicity and/or complications from the intervention and at least two weeks have to pass prior to starting therapy.
- 3.2.10 Has known additional malignancy that progressed or required treatment within last 5 years. Exceptions include basal cell carcinoma of the skin or squamous cell carcinoma of the skin that has undergone potentially curative therapy or in situ cervical cancer that has been adequately treated.
- 3.2.11 Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. Subjects with previously treated brain metastases may participate provided they are stable (without evidence of progression by imaging for at least 28 days prior to registration and any neurologic symptoms have returned to baseline), have no evidence of new or enlarging brain metastases, and are not using steroids for at least 7 days prior to registration. This exception does not include known carcinomatous meningitis which is excluded regardless of clinical stability.
Note: steroids as premedication are permitted in this study. Refer section 3.2.4
- 3.2.12 Has active autoimmune disease that has required systemic treatment in the past 2 years (i.e. with use of disease modifying agents, corticosteroids or immunosuppressive drugs). Replacement therapy (eg. thyroxine, insulin, or physiologic corticosteroid replacement therapy for adrenal or pituitary insufficiency, etc.) is not considered a form of systemic treatment.
- 3.2.13 Patients who have evidence of active, noninfectious pneumonitis/interstitial lung disease or have a history of severe pneumonitis/interstitial lung disease that required treatment with steroids are not eligible for this study.

- 3.2.14 Has an active infection requiring systemic therapy.
- 3.2.15 Has a history or current evidence of any condition, therapy, or laboratory abnormality that might confound the results of the trial, interfere with the subject's participation for the full duration of the trial, or is not in the best interest of the subject to participate, in the opinion of the treating investigator.
- 3.2.16 Has known psychiatric or substance abuse disorders that would interfere with cooperation with the requirements of the trial.
- 3.2.17 Is pregnant or breastfeeding, or expecting to conceive or father children within the projected duration of the trial, starting with the pre-screening or screening visit through 120 days after the last dose of trial treatment.
- 3.2.18 Has received prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent or with an agent directed to another stimulatory or co-inhibitory T-cell receptor (e.g., CTLA-4, OX 40, CD137).
- 3.2.19 Has a known history of Human Immunodeficiency Virus (HIV) (HIV 1/2 antibodies).
- 3.2.20 *Has received a live vaccine or live attenuated vaccine within 30 days of planned start of study therapy. Administration of killed vaccines is allowed.*

Note: Seasonal influenza vaccines for injection are generally inactivated flu vaccines and are allowed; however intranasal influenza vaccines (e.g., Flu-Mist®) are live attenuated vaccines, and are not allowed.

4 TREATMENT PLAN

4.1 Treatment Overview

This is a phase II, prospective trial testing the combination of pembrolizumab and carboplatin in patients with HER-2 negative, CTCs positive (defined as CTCs $\geq$ 5) MBC. Patients will be treated with pembrolizumab 200mg intravenously (IV) on Day 1 of each cycle (1 cycle = 21 days) with carboplatin AUC 5 IV on Day 1 beginning at cycle 1. Treatments continue until patient completes 24 months of treatment 35 administrations of study medication, whichever is later, or until progressive disease or unacceptable toxicity. The primary hypothesis will be that the combination of pembrolizumab IV and carboplatin IV will increase PFS in patients with $\geq$ 5 CTCs compared with the historical median of 4.9 months.

4.2 Treatment Administration

The treatment to be used in this trial is outlined below in Table 2.

Table 2. Trial Treatment

Drug	Premedications	Dose	Dose Frequency	Route of Administration	Regimen/Treatment Period	Use
Pembrolizumab	n/a	200 mg	Q3W	IV infusion	Day 1 of each 3 week cycle	Experimental
Carboplatin	Routine premedication per institutional standard*	AUC5	Q3W	IV infusion	Day 1 of each 3 week cycle	Standard

* steroids as pre-medication for carboplatin is permitted

Note: Trial treatment should begin as close as possible to the date of registration.

4.2.1 *Pembrolizumab*

Treatment should be administered on Day 1 of each cycle, but may be administered up to 3 days before or after the scheduled Day 1 of each cycle due to administrative reasons. All trial treatments will be administered on an outpatient basis.

Pembrolizumab 200 mg will be administered as a 30 minute IV infusion on Day 1 of each cycle. Every effort should be made to target infusion timing to be as close to 30 minutes as possible. However, given the variability of infusion pumps from site to site, a window of -5 minutes and +10 minutes is permitted (i.e., infusion time is 30 minutes: -5 min/+10 min).

The Pharmacy Manual contains specific instructions for the preparation of the pembrolizumab infusion fluid and administration of infusion solution.

4.2.2 *Carboplatin*

Carboplatin will be given on Day 1 (± 3 days) of each cycle. Pembrolizumab will be administered first, followed by administration of carboplatin. There are no time requirements between infusions.

Carboplatin should be given as a 30 to 60 min i.v. infusion to an AUC of 5 on day 1, with the dose calculated according to the institutional standard.

4.3 Dose Modification

4.3.1 *Pembrolizumab Dose Delay & Modifications*

AEs (both non-serious and serious) associated with pembrolizumab exposure may represent an immunologic etiology. These AEs may occur shortly after the first dose or several months after the last dose of treatment. Pembrolizumab must be withheld for drug-related toxicities and severe or life-threatening AEs as per Table 3 below. See Section 4.5 for supportive care guidelines, including use of corticosteroids. Patients who are removed from treatment with pembrolizumab due to disease progression or unacceptable toxicity will come off the study treatment and begin the safety follow up phase. Carboplatin must be withheld for drug-related toxicities and severe or life-threatening AEs as per Table 5 below. Patients who experience unacceptable toxicity due to carboplatin may continue treatment with single agent pembrolizumab.

Any patient who receives at least one dose of study therapy will be evaluable for toxicity endpoints. Toxicity will be assessed according to NCI's CTCAE, version 4.03 (see Appendix 12.2).

Table 3. Dose Modification Guidelines for Pembrolizumab-Related AEs

General instructions: 1. Corticosteroid taper should be initiated upon AE improving to Grade 1 or less and continue to taper over at least 4 weeks. 2. For situations where pembrolizumab has been withheld, pembrolizumab can be resumed after AE has been reduced to Grade 1 or 0 and corticosteroid has been tapered. Pembrolizumab should be permanently discontinued if AE does not resolve within 12 weeks of last dose or corticosteroids cannot be reduced to ≤ 10 mg prednisone or equivalent per day within 12 weeks. 3. For severe and life-threatening irAEs, IV corticosteroid should be initiated first followed by oral steroid. Other immunosuppressive treatment should be initiated if irAEs cannot be controlled by corticosteroids.				
Immune-related AEs	Toxicity grade or conditions (CTCAEv4.0)	Action taken to pembrolizumab	irAE management with corticosteroid and/or other therapies	Monitor and follow-up
Pneumonitis	Grade 2	Withhold	• Administer corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent) followed by taper	• Monitor participants for signs and symptoms of pneumonitis • Evaluate participants with suspected pneumonitis with radiographic imaging and initiate corticosteroid treatment • Add prophylactic antibiotics for opportunistic infections
	Grade 3 or 4, or recurrent Grade 2	Permanently discontinue		
Diarrhea / Colitis	Grade 2 or 3	Withhold	• Administer corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent) followed by taper	• Monitor participants for signs and symptoms of enterocolitis (ie, diarrhea, abdominal pain, blood or mucus in stool with or without fever) and of bowel perforation (ie, peritoneal signs and ileus). • Participants with $\geq$ Grade 2 diarrhea suspecting colitis should consider GI consultation and performing endoscopy to rule out colitis. • Participants with diarrhea/colitis should be advised to drink liberal quantities of clear fluids. If sufficient oral fluid intake is not feasible, fluid and electrolytes should be substituted via IV infusion.
	Grade 4	Permanently discontinue		
AST / ALT elevation or Increased bilirubin	Grade 2	Withhold	• Administer corticosteroids (initial dose of 0.5- 1 mg/kg prednisone or equivalent) followed by taper	• Monitor with liver function tests (consider weekly or more frequently until liver enzyme value returned to baseline or is stable)

	Grade 3 or 4	Permanently discontinue	Administer corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent) followed by taper	
Type 1 diabetes mellitus (T1DM) or Hyperglycemia	Newly onset T1DM or Grade 3 or 4 hyperglycemia associated with evidence of β -cell failure	Withhold	- Initiate insulin replacement therapy for participants with T1DM - Administer anti-hyperglycemic in participants with hyperglycemia	Monitor participants for hyperglycemia or other signs and symptoms of diabetes.
Hypophysitis	Grade 2	Withhold	Administer corticosteroids and initiate hormonal replacements as clinically indicated.	Monitor for signs and symptoms of hypophysitis (including hypopituitarism and adrenal insufficiency)
	Grade 3 or 4	Withhold or permanently discontinue ¹		
Hyperthyroidism	Grade 2	Continue	Treat with non-selective beta-blockers (eg, propranolol) or thionamides as appropriate	Monitor for signs and symptoms of thyroid disorders.
	Grade 3 or 4	Withhold or permanently discontinue ¹		
Hypothyroidism	Grade 2-4	Continue	Initiate thyroid replacement hormones (eg, levothyroxine or liothyronine) per standard of care	Monitor for signs and symptoms of thyroid disorders.
Nephritis and Renal dysfunction	Grade 2	Withhold	Administer corticosteroids (prednisone 1-2 mg/kg or equivalent) followed by taper.	Monitor changes of renal function
	Grade 3 or 4	Permanently discontinue		
Myocarditis	Grade 1 or 2	Withhold	Based on severity of AE administer corticosteroids	Ensure adequate evaluation to confirm etiology and/or exclude other causes
	Grade 3 or 4	Permanently discontinue		
All other immune-related AEs	Intolerable/persistent Grade 2	Withhold	Based on type and severity of AE administer corticosteroids	Ensure adequate evaluation to confirm etiology and/or exclude other causes
	Grade 3	Withhold or discontinue based on the type of event. Events that require discontinuation include and not limited to:		

		Gullain-Barre Syndrome, encephalitis		
	Grade 4 or recurrent Grade 3	Permanently discontinue		
1. Withhold or permanently discontinue pembrolizumab is at the discretion of the investigator or treating physician. NOTE: For participants with Grade 3 or 4 immune-related endocrinopathy where withhold of pembrolizumab is required, pembrolizumab may be resumed when AE resolves to $\leq$ Grade 2 and is controlled with hormonal replacement therapy or achieved metabolic control (in case of T1DM).				

4.3.2 Carboplatin Dose Delays & Modifications

Starting dose and dose modifications for unacceptable toxicity are listed in Table 4. For toxicity that is thought to be related to carboplatin, the dose of carboplatin will be modified according to the schedule displayed in Table 5.

Dose adjustments are to be made according to the greatest degree of toxicity.

Table 4. Carboplatin Doses

Dose Level	Carboplatin (AUC)
0	5
-1	4

Table 5. Dose Reduction Criteria & Guidelines for Management of Carboplatin-Associated Toxicity**

Toxicity	Occurrence	Guidelines for Management	Dose Modification
ANC < 500 cells/mm³ (nadir count) with neutropenic fever >38°C OR Delay of next cycle due to persistent neutropenia (ANC < 1500 cells/mm³) OR Neutropenia < 500 cells/mm³ for > 1 week	1 st occurrence	Give G-CSF as outlined below in Section 4.3.1.1	Hold* carboplatin until recovery to ANC $\geq$ 1.5x10 ⁹ cells/L and platelets > 100x10 ⁹ cells/L and restart at the same dose level.
	Additional occurrences	Give G-CSF as outlined below in Section 4.3.1.1	Hold* until recovery to ANC $\geq$ 1.5x10 ⁹ cells/L and platelets > 100x10 ⁹ cells/L. After recovery, reduce by 1 dose level for subsequent cycles (AUC 4).
Thrombocytopenia Grade 3 or 4	1 st occurrence	None	Hold* carboplatin for that cycle and then reduce dose to next lower level (AUC 4).
	Additional occurrences	None	Discontinue treatment
Grade 2 peripheral neuropathy**	Any	Gabapentin or similar medications can be given at the discretion	Hold* until improved to $\leq$ grade 1 and reduce by 1 dose level for

		of the treating physician.	subsequent cycles (AUC 4).
Grade 3 or higher peripheral neuropathy**			Discontinue treatment
Other Toxicity $\leq$ Grade 2	Any	Manage symptomatically, if possible.	If unacceptable to patient or medically concerning, may hold* until recovery to $\leq$ grade 1 or baseline and then restart at the same dose level.
Other Toxicity $\geq$ Grade 3 (excluding anemia)	Any	Treat as medically appropriate	Hold* until resolved to $\leq$ grade 1 or baseline and then dose reduce by 1 dose level for subsequent cycles (AUC 4).

*Doses of carboplatin that are held will be missed and will not be made up for that cycle. Carboplatin should be given at the next planned cycle.

**In the event of peripheral neuropathy, the investigator can decide whether to hold and dose reduce carboplatin.

++Determination as to whether or not toxicity is related to carboplatin will be made by the treating physician. Please refer to Section 5.4.3 for guidelines determining the attribution of myelosuppression. Toxicities based on lab results will be assessed according to labs drawn at treatment days only (days 1 and 8) unless the treating physician feels that it would be otherwise unsafe.

4.3.2.1 G-CSF Administration

If needed, Neulasta can be given on day 9 of a cycle. Neupogen can also be given at the discretion of the treating investigator.

4.3.2.2 Dose Modifications Due to Myelosuppression

In the event of Grade 3 or higher myelosuppression the dose of carboplatin will be held and subsequently reduced (as outlined in 4.3.1 above). Dose modifications will be based only on results of labs drawn at the day of treatment unless the treating physician feels it would be unsafe to proceed with treatment based on other interim results.

Dosing interruptions are permitted in the case of medical / surgical events or logistical reasons not related to study therapy (e.g., elective surgery, unrelated medical events, patient vacation, and/or holidays). Subjects should be placed back on study therapy within 3 weeks of the scheduled interruption, unless otherwise discussed with the PI and Quality Assurance Monitor (QAM). The reason for interruption should be documented in the patient's study record.

4.4 Concomitant Medications/Vaccinations (allowed & prohibited)

Medications or vaccinations specifically prohibited in the exclusion criteria are not allowed during the ongoing trial. If there is a clinical indication for one of these or other medications or vaccinations specifically prohibited during the trial, discontinuation from trial therapy or vaccination may be required. The investigator should discuss any questions regarding this with the Merck Clinical team. The final decision on any supportive therapy or vaccination rests with the investigator and/or the subject's primary physician.

4.4.1 Acceptable Concomitant Medications

All treatments that the investigator considers necessary for a subject's welfare may be administered at the discretion of the investigator in keeping with the community standards of medical care. All concomitant medication will be recorded on the electronic case report

form (eCRF) including all prescription, over-the-counter (OTC), herbal supplements, and IV medications and fluids. If changes occur during the trial period, documentation of drug dosage, frequency, route, and date may also be included on the eCRF.

All concomitant medications received within 28 days before the first dose of trial treatment and 30 days after the last dose of trial treatment should be recorded. Concomitant medications administered after 30 days after the last dose of trial treatment for the purpose of managing treatment-related toxicity should be recorded for SAEs and events of clinical interest (ECIs) as defined in Section 8.4.3. The NU SAE form will be used for reporting these.

4.4.2 *Prohibited Concomitant Medications*

Subjects are prohibited from receiving the following therapies during the Screening and Treatment Phase (including retreatment for post-complete response relapse) of this trial:

- Biological therapy
- Immunotherapy not specified in this protocol
- Chemotherapy not specified in this protocol
- Investigational agents other than pembrolizumab
- Radiation therapy
 - Note: Radiation therapy to a symptomatic solitary lesion or to the brain may be allowed at the treating investigator's discretion. Radiated lesions cannot be used as a target lesion for disease assessment
- Live vaccines within 30 days prior to the first dose of trial treatment and while participating in the trial. Examples of live vaccines include, but are not limited to, the following: measles, mumps, rubella, varicella/zoster, yellow fever, rabies, BCG, and typhoid vaccine.
- Systemic glucocorticoids for any purpose other than to modulate symptoms from an ECI of suspected immunologic etiology. The use of physiologic doses of corticosteroids may be approved after consultation with PI.

Note: steroids as pre-medication are permitted in this study.

Subjects who, in the assessment by the investigator, require the use of any of the aforementioned treatments for clinical management should be removed from the trial. Subjects may receive other medications that the investigator deems to be medically necessary.

The Exclusion Criteria describes other medications which are prohibited in this trial.

There are no prohibited therapies during the Post-Treatment Follow-up Phase.

4.5 **Rescue Medications & Supportive Care**

Subjects should receive appropriate supportive care measures as deemed necessary by the treating investigator. Suggested supportive care measures for the management of AEs with potential immunologic etiology are outlined below. Where appropriate, these guidelines include the use of oral or intravenous treatment with corticosteroids as well as additional anti-inflammatory agents if symptoms do not improve with administration of corticosteroids. Note that several courses of steroid tapering may be necessary as symptoms may worsen when the steroid dose is decreased. For each disorder, attempts should be made to rule out other causes such as metastatic disease or bacterial or viral infection, which might require additional supportive care. The treatment guidelines are intended to be applied when the investigator determines the events to be related to pembrolizumab.

Note: if after the evaluation the event is determined not to be related, the investigator is instructed to follow the ECI reporting guidance but does not need to follow the treatment guidance. Refer to Section 4.2 for dose modification.

It may be necessary to perform conditional procedures such as bronchoscopy, endoscopy, or skin photography as part of evaluation of the event.

Management of Infusion Reactions: Signs and symptoms usually develop during or shortly after drug infusion and generally resolve completely within 24 hours of completion of infusion.

Table 6 below shows treatment guidelines for subjects who experience an infusion reaction associated with administration of pembrolizumab (MK-3475).

Table 6 Infusion Reaction Treatment Guidelines

NCI CTCAE Grade	Treatment	Premedication at subsequent dosing
Grade 1 Mild reaction; infusion interruption not indicated; intervention not indicated	Increase monitoring of vital signs as medically indicated until the subject is deemed medically stable in the opinion of the investigator.	None
Grade 2 Requires infusion interruption but responds promptly to symptomatic treatment (e.g., antihistamines, NSAIDs, narcotics, IV fluids); prophylactic medications indicated for < =24 hrs	Stop Infusion and monitor symptoms. Additional appropriate medical therapy may include but is not limited to: IV fluids Antihistamines NSAIDs Acetaminophen Narcotics Increase monitoring of vital signs as medically indicated until the subject is deemed medically stable in the opinion of the investigator. If symptoms resolve within one hour of stopping drug infusion, the infusion may be restarted at 50% of the original infusion rate (e.g., from 100 mL/hr to 50 mL/hr). Otherwise dosing will be held until symptoms resolve and the subject should be premedicated for the next scheduled dose. Subjects who develop Grade 2 toxicity despite adequate premedication should be permanently discontinued from further trial treatment administration.	Subject may be premedicated 1.5h (± 30 minutes) prior to infusion of pembrolizumab (MK-3475) with: Diphenhydramine 50 mg po (or equivalent dose of antihistamine). Acetaminophen 500-1000 mg po (or equivalent dose of antipyretic).
Grades 3 or 4 Grade 3: Prolonged (i.e., not rapidly responsive to symptomatic medication and/or brief interruption of infusion); recurrence of symptoms following initial improvement; hospitalization indicated for other clinical sequelae (e.g., renal impairment, pulmonary infiltrates) Grade 4: Life-threatening; pressor or ventilatory support indicated	Stop Infusion. Additional appropriate medical therapy may include but is not limited to: IV fluids Antihistamines NSAIDs Acetaminophen Narcotics Oxygen Pressors Corticosteroids Epinephrine Increase monitoring of vital signs as medically indicated until the subject is deemed medically stable in the opinion of the investigator. Hospitalization may be indicated. Subject is permanently discontinued from further trial treatment administration.	No subsequent dosing
Appropriate resuscitation equipment should be available in the room and a physician readily available during the period of drug administration.		

4.6 Diet/Activity/Contraception/Other Considerations

Subjects should maintain a normal diet unless modifications are required to manage an AE such as diarrhea, nausea or vomiting.

Pembrolizumab may have adverse effects on a fetus in utero. Furthermore, it is not known if pembrolizumab has transient adverse effects on the composition of sperm. Non-pregnant, non-breast-feeding women may be enrolled if they are willing to use 2 methods of birth control or are considered highly unlikely to conceive. Highly unlikely to conceive is defined as 1) surgically sterilized, or 2) postmenopausal (a woman who is ≥ 45 years of age and has not had menses for greater than 1 year will be considered postmenopausal), or 3) not heterosexually active for the duration of the study. The two birth control methods can be either two barrier methods or a barrier method plus a hormonal method to prevent pregnancy. Subjects should start using birth control from study Visit 1 throughout the study period up to 120 days after the last dose of study therapy.

The following are considered adequate barrier methods of contraception: diaphragm, condom (by the partner), copper intrauterine device, sponge, or spermicide. Appropriate hormonal contraceptives will include any registered and marketed contraceptive agent that contains an estrogen and/or a progestational agent (including oral, subcutaneous, intrauterine, or intramuscular agents).

Subjects should be informed that taking the study medication may involve unknown risks to the fetus (unborn baby) if pregnancy were to occur during the study. In order to participate in the study they must adhere to the contraception requirement (described above) for the duration of the study and during the follow-up period defined in section 8.4.2-Reporting of Pregnancy and Lactation to the Sponsor and to Merck. If there is any question that a subject will not reliably comply with the requirements for contraception, that subject should not be entered into the study.

If a subject inadvertently becomes pregnant while on treatment with pembrolizumab, the subject will immediately be removed from the study. The site will contact the subject at least monthly and document the subject's status until the pregnancy has been completed or terminated. The outcome of the pregnancy will be reported to the PI and to Merck without delay and within 24 hours to the Sponsor and within 2 working days to Merck if the outcome is a serious adverse experience (e.g., death, abortion, congenital anomaly, or other disabling or life-threatening complication to the mother or newborn).

The study investigator will make every effort to obtain permission to follow the outcome of the pregnancy and report the condition of the fetus or newborn to the Sponsor. If a male subject impregnates his female partner the study personnel at the site must be informed immediately and the pregnancy reported to the Sponsor and to Merck and followed as described above and in Section 8.4.2.

It is unknown whether pembrolizumab is excreted in human milk. Since many drugs are excreted in human milk, and because of the potential for serious adverse reactions in the nursing infant, subjects who are breast-feeding are not eligible for enrollment.

4.7 Subject Withdrawal/Discontinuation Criteria from Treatment and/or Overall Study

Subjects may withdraw consent at any time for any reason or be dropped from the trial at the discretion of the investigator should any untoward effect occur. In addition, a subject may be withdrawn if enrollment into the trial is inappropriate, the trial plan is violated, or for administrative and/or other safety reasons. Specific details regarding discontinuation or withdrawal are provided below. The reason(s) for discontinuation must be clearly documented on the appropriate eCRF and may include:

A subject must be discontinued from the trial for any of the following reasons:

- Patient voluntarily withdraws from treatment (follow-up permitted)

- The subject or legal representative (such as a parent or legal guardian) withdraws consent (no follow-up permitted).
- Confirmed radiographic disease progression
Note: For unconfirmed radiographic disease progression, please see Section 5.2.2
- Unacceptable adverse events
- Intercurrent illness that prevents further administration of treatment
- Investigator's decision to withdraw the subject
- The subject has a confirmed positive serum pregnancy test
- Noncompliance with trial treatment or procedure requirements
- The subject is lost to follow-up
- Completed 24 months of uninterrupted treatment with pembrolizumab or 35 administrations of study medication, whichever is later.
- Administrative reasons

The End of Treatment and Follow-up visit procedures are listed in Section 5 (Study Flow Chart). After the end of treatment, each subject will be followed for 30 days for AEs monitoring (serious adverse events (SAEs) will be collected for 90 days after the end of treatment as described in Section 8.2.3). Subjects who discontinue for reasons other than progressive disease will have post-treatment follow-up for disease status until disease progression, initiating a non-study cancer treatment, withdrawing consent or becoming lost to follow-up. After documented disease progression each subject will be followed by telephone for overall survival until death, withdrawal of consent, or the end of the study, whichever occurs first.

4.8 Withdrawal/Discontinuation

When a subject discontinues/withdraws prior to trial completion, all applicable activities scheduled for the final trial visit should be performed at the time of discontinuation. Any AEs which are present at the time of discontinuation/withdrawal should be followed in accordance with the safety requirements outlined in Section 8.0 – Adverse Events. Subjects who a) attain a CR or b) complete 24 months of treatment (or 35 administrations) with pembrolizumab may discontinue treatment with the option of restarting treatment if they meet the criteria specified in Section 3. After discontinuing treatment following assessment of CR these subjects should return to the site for a Safety Follow-up Visit, including an end of treatment scan, and then proceed to the Follow-Up Period of the study.

4.8.1 Discontinuation of Study Therapy after CR

Discontinuation of treatment may be considered for subjects who have attained a confirmed CR that have been treated for at least 24 weeks with pembrolizumab and had at least two treatments with pembrolizumab beyond the date when the initial CR was declared.

4.9 Duration of Therapy

Patients will continue to receive 24 months of uninterrupted treatment with pembrolizumab or 35 administrations of study medication (whichever is later) until any of the following occur:

- Confirmed disease progression
- Development of an inter-current illness that prevents further administration of treatment
- Unacceptable AEs
- Patient decides to withdraw from either study treatment or the as a whole study
- The treating investigator determines that the patient should be taken off treatment for any reason (i.e. changes in condition, inability to comply with study treatment or procedures)
- Subjects who attain a CR may discontinue treatment and should return to the site for a Safety Follow-up Visit and then proceed to the Follow-Up Period of the study.

4.10 Duration of Follow Up

4.10.1 *Safety Follow-Up Visit*

The mandatory Safety Follow-Up Visit should be conducted within 30 days after the last dose of trial treatment or before the initiation of a new anti-cancer treatment, whichever comes first. All AEs that occur prior to the Safety Follow-Up Visit should be recorded. Subjects with an AE of Grade > 1 will be followed until the resolution of the AE to Grade 0-1 or until the beginning of a new anti-neoplastic therapy, whichever occurs first. SAEs that occur within 90 days of the end of treatment or before initiation of a new anti-cancer treatment should also be followed and recorded.

4.10.2 *Follow-up Visits*

Subjects who discontinue trial treatment for a reason other than disease progression will move into the Follow-Up Phase and should be assessed every 9 weeks (63 ± 7 days) by radiologic imaging to monitor disease status. After 1 year, the imaging time point will occur every 12 weeks (± 7 days). Every effort should be made to collect information regarding disease status until the start of new anti-neoplastic therapy, disease progression, death, end of the study or if the subject begins retreatment with pembrolizumab. Information regarding post-study anti-neoplastic treatment will be collected if new treatment is initiated.

4.10.3 *Survival Follow-up*

Once a subject experiences confirmed disease progression or starts a new anti-cancer therapy, the subject moves into the survival follow-up phase and should be contacted by telephone every 12 weeks to assess for survival status until death, withdrawal of consent, or the end of the study, whichever occurs first. Information regarding the subject's next line of therapy and clinical response will be collected if a new treatment is initiated while the patient is on follow up.

4.11 Patient replacement

If a patient comes off study before the first dose of study treatment, the patient may be replaced.

5 STUDY PROCEDURES

5.1 Study Flow Chart

Trial Period:	Screening Phase		Treatment Cycles ¹								EOT	Post-Treatment		
Treatment Cycle/Title:	Pre-screening (Visit 1)	Main Study Screening (Visit 2)	1	2	3	4	To be repeated beyond 8 cycles ³				Discontinuation	Safety Follow-up	Follow Up Visits ²	Survival Follow-Up ⁴
							5	6	7	8				
Scheduling Window (Days): 14		-28 to -1	± 3	± 3	± 3	± 3	± 3	± 3	± 3	± 3	At time of Discontinuation	30 days post discontinuation	Every 9 weeks post discontinuation	Every 12 weeks
Administrative Procedures														
Informed Consent		X												
Demographics and Medical History		X												
Prior and Concomitant Medication Review	X	X	X	X	X	X	X	X	X	X	X	X	X	
Pembrolizumab 200mg			X	X	X	X	X	X	X	X				
Carboplatin AUC 5			X	X	X	X	X	X	X	X				
Post-study anticancer therapy status												X	X	X
Survival Status											X	X	X	X
Clinical Procedures/Assessments														
Review AEs/SAEs ⁵		X	X	X	X	X	X	X	X	X	X	X	X	
Laboratory review ⁶		X	X											
Physical Exam, Vital Signs and Weight		X	X	X	X	X	X	X	X	X	X			
ECOG Performance Status		X	X	X	X	X	X	X	X	X	X			
Lesion Photographs ⁷		Optional: photos may be taken during study treatment per treating physician discretion.												
Laboratory Procedures/Assessments: analysis performed by LOCAL laboratory														
Pregnancy Test – Urine or Serum β-HCG ⁸		X												
PT/INR and aPTT		X												
CBC with Differential ⁹		X	X	X	X	X	X	X	X	X	X			
Comprehensive Serum Chemistry Panel ¹⁰		X	X	X	X	X	X	X	X	X	X			
Urinalysis ¹¹		X												
TSH and reflexive T3,T4 ¹²		X		X		X		X		X	X			
Efficacy Measurements														
Tumor Imaging ¹³		X			X			X			X ¹⁴		X ²	
Tumor Biopsies/Archival Tissue Collection/Correlative Studies Blood														
Archival Tissue Collection ¹⁵		X												
Correlative Studies Blood Collection ¹⁶		X			X	X					X			
CTC Enumeration ¹⁷	X													

Trial Period:	Screening Phase	Treatment Cycles ¹	EOT	Post-Treatment
- Treatment will be given on Day 1 of each 21 day cycle (± 3 days). Trial treatment should begin as close as possible to the date on which treatment is allocated/assigned. - Subjects who discontinue trial treatment for a reason other than disease progression will move into the Follow-Up Phase and should be assessed every 9 weeks (63 ± 7 days) by radiologic imaging to monitor disease status. After 1 year, the imaging time point will occur every 12 weeks (± 7 days). - Treatment cycles continue until the patient completes 24 months of treatment or 35 administrations of study medication, whichever is later. - Once a subject experiences confirmed disease progression or starts a new anti-cancer therapy, the subject moves into the survival follow-up phase and should be contacted by telephone every 12 weeks to assess for survival status until death, withdrawal of consent, or the end of the study, whichever occurs first. <i>Note: Information regarding the subject's next line of therapy and clinical response will be collected if a new treatment is initiated while the patient is on follow up.</i> - Pembrolizumab must be withheld for drug-related toxicities and severe or life-threatening AEs as per Table 3. Carboplatin must be withheld for drug-related toxicities and severe or life-threatening AEs per Table 5. AE/SAE to be followed up until resolved or patient is off study - Labs within 4 days of C1D1 (or on Day 1 of treatment) must be drawn and reviewed by Investigator prior to treatment. - Optional photos for disease assessments may be taken by study staff on a study-supplied digital camera per treating physician discretion. It is preferred that these photos be taken on a white background (wall of patient room) using a ruler measurement when possible. Photographs should not include any identifying characteristics. - Perform on women of childbearing potential only. FOCBP should have a negative urine or serum pregnancy test within 7 days prior to registration; and this must be repeated within 3 days (72 hours) prior to first dose of study drug if urine pregnancy results cannot be confirmed as negative, a serum pregnancy test will be required. - CBC with Diff will include absolute lymphocyte count, absolute neutrophil count, red blood cell count, WBC (total and differential) platelet count, hemoglobin, and hematocrit. - Serum Chemistry Panel will include calcium, chloride, magnesium, sodium, potassium, blood urea nitrogen, bicarbonate, phosphorus, glucose, total bilirubin, aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), albumin, total protein. Additional chemistry labs will include: lactate dehydrogenase LDH, uric acid. If total bilirubin is $\geq 2 \times \text{ULN}$ (and no evidence of Gilbert's syndrome) then fractionate into direct bilirubin. - Urinalysis will include blood, glucose, specific gravity and a microscopic exam (if abnormal). - Thyroid function will be evaluated at baseline and every 2 cycles thereafter - Tumor imaging will occur every 9 weeks following initiation of treatment. CT c/a/p AND bone scan. In place of bone scan, a PET/CT may be used. If CT is contraindicated, MRI may be used. Any additional imaging such as MRI can be completed at the physician's discretion. The same modality should be used throughout the study. <i>Note: Bone scans/PET are to be repeated at each time point ONLY if the patient has bone metastasis or if it is clinically indicated. If the original bone scan/PET during screening is negative, then bone scans/PET are not required in subsequent visits.</i> - Subjects receive an end of treatment scan when they discontinue treatment with pembrolizumab due to unacceptable toxicity. - The archived tumor tissue specimens may be from prior surgery or from prior diagnostic biopsy (second choice) of primary or metastatic tumor specimen (first choice). Unavailability of archived tissue will not render subject ineligible for study. - After enrollment patients will have a) CTCs PDI-1 (CellSearch), CTCs PDL1 and CAMLs (CellSieve), and T-cell receptor NGS (tissue at baseline/blood baseline and follow-up). - CTC enumeration by CellSearch is a screening test to identify candidate patients and is standard of care. Abbreviations: Adverse event (AE); serious adverse event (SAE), end of treatment (EOT)				

5.2 Trial Procedures

The Study Flow Chart - Section 5.1 summarizes the trial procedures to be performed at each visit. Individual trial procedures are described in detail below. It may be necessary to perform these procedures at unscheduled time points if deemed clinically necessary by the investigator.

5.2.1 *Administrative Procedures*

Informed Consent

The Investigator must obtain documented consent from each potential subject prior to participating in a clinical trial.

General Informed Consent

Consent must be documented by the subject's dated signature or by the subject's legally acceptable representative's dated signature on a consent form along with the dated signature of the person conducting the consent discussion. A copy of the signed and dated consent form should be given to the subject before participation in the trial.

The initial informed consent form, any subsequent revised written informed consent form and any written information provided to the subject must receive IRB approval in advance of use. The subject or his/her legally acceptable representative should be informed in a timely manner if new information becomes available that may be relevant to the subject's willingness to continue participation in the trial. The communication of this information will be provided and documented via a revised consent form or addendum to the original consent form that captures the subject's dated signature or by the subject's legally acceptable representative's dated signature.

Specifics about a trial and the trial population will be added to the consent form template at the protocol level.

The informed consent will adhere to IRB requirements, applicable laws and regulations and Sponsor requirements.

Inclusion/Exclusion Criteria

All inclusion and exclusion criteria will be reviewed by the investigator or qualified designee to ensure that the subject qualifies for the trial.

Medical History

A medical history will be obtained by the investigator or qualified designee. Medical history will include all active conditions, and any condition diagnosed within the prior 10 years that are considered to be clinically significant by the Investigator. Details regarding the disease for which the subject has enrolled in this study will be recorded separately and not listed as medical history.

Prior and Concomitant Medications Review

Prior Medications

The investigator or qualified designee will review prior medication use, including any protocol-specified washout requirement, and record prior medication taken by the subject within 28 days before starting the trial. Treatment for the disease for which the subject has enrolled in this study will be recorded separately and not listed as a prior medication.

Concomitant Medications

The investigator or qualified designee will record medication, if any, taken by the subject during the trial. All medications related to reportable SAEs and ECIs should be recorded as defined in Section 8.0.

Disease Details and Treatments

Disease Details

The investigator or qualified designee will obtain prior and current details regarding disease status.

Prior Treatment Details

The investigator or qualified designee will review all prior cancer treatments including systemic treatments, radiation and surgeries.

Subsequent Anti-Cancer Therapy Status

The investigator or qualified designee will review all new anti-neoplastic therapy initiated after the last dose of trial treatment. If a subject initiates a new anti-cancer therapy within 30 days after the last dose of trial treatment, the 30 day Safety Follow-up visit must occur before the first dose of the new therapy. Once new anti-cancer therapy has been initiated the subject will move into survival follow-up.

Clinical Procedures/Assessments

Adverse Event (AE) Monitoring

Please refer to section 8.0 for detailed information regarding the assessment and recording of AEs.

Full Physical Exam

The investigator or qualified designee will perform a complete physical exam during the screening period. Clinically significant abnormal findings should be recorded as medical history. A full physical exam should be performed during screening,

Directed Physical Exam

For cycles that do not require a full physical exam per the Trial Flow Chart, the investigator or qualified designee will perform a directed physical exam as clinically indicated prior to trial treatment administration.

Vital Signs

The investigator or qualified designee will take vital signs at screening, prior to the administration of each dose of trial treatment and at treatment discontinuation as specified in the Study Flow Chart (Section 5.1). Vital signs should include temperature, pulse, weight and blood pressure. Height will be measured at screening only.

Eastern Cooperative Oncology Group (ECOG) Performance Scale

The investigator or qualified designee will assess ECOG status (see Section 12.1) at screening, prior to the administration of each dose of trial treatment and discontinuation of trial treatment as specified in the Trial Flow Chart.

Tumor Imaging and Assessment of Disease

Tumor Tissue Collection and Correlative Studies Blood Sampling

Patients will be evaluated by treating physician with standard imaging (CT chest/abdomen/pelvis and bone scan, or PET/CT) every 9 weeks beginning at the time of initiation of treatment.

Laboratory Procedures/Assessments

Details regarding specific laboratory procedures/assessments to be performed in this trial are provided in the Lab Manual.

Laboratory Safety Evaluations (Hematology, Chemistry and Urinalysis)

Laboratory tests for hematology, chemistry, urinalysis, and others are specified in Study Flow Chart.

6 ENDPOINT ASSESSMENT

6.1 Summary

This study is a prospective, single arm Phase II study with a primary endpoint PFS. Median survival will be compared with a historical median survival of 4.9 months that is similar in first-line recurrence or de-novo stage IV TNBC. Unless otherwise noted, all analysis will be performed using SAS 9.4 (SAS/STAT 13.2) or later.

6.2 Sample Size Justification

Sample size calculations were performed using the method of Lawless as implemented in the SWOG online calculator (http://www.swogstat.org/stat/public/one_survival.htm). The length of accrual was assumed to be 18 months followed by an additional 18 months of follow-up for a total study period of 3 years. To have 90% power to detect an improvement in median PFS to 8.9 months assuming a two-sided alpha of 0.05, this Phase II study requires 32 treated and evaluable subjects. Allowing for 10% dropout, we will treat a maximum of 38 subjects. Assuming 35-40% of screened subjects will have ≥ 5 CTCs, we will screen approximately 100 subjects.

6.3 Primary Endpoint

To test if the combination of Carboplatin intravenously and pembrolizumab will increase PFS in patients with ≥ 5 CTCs compared with the historical median of 4.9 months.

PFS will be based on investigator assessment. PFS will be defined as the time from the first study treatment to the first occurrence of progression or death, and will be assessed every 9 weeks after beginning of treatment.

Any patient who has completed at least 3 cycles of study treatment are evaluable for this endpoint.

6.4 Secondary Endpoints

- To determine the toxicity and safety of the combination pembrolizumab-carboplatin in patients with CTC positive, HER2 negative MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV. Adverse Events will be assessed according to the NCI CTCAE 4.03.
- To evaluate the impact on OS, ORR and CBR with combination Carboplatin - pembrolizumab in patients with CTCs positive MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuvant therapy or are de-novo stage IV. ORR and CBR will be evaluated according to RECIST criteria and in relation to PDL-1 expression in tissue and CTCs.
- To assess immune-related response using tumor response by irRECIST as immune-related PR (irPR) or CR (irCR). We will define immune-related clinical benefit rate as immune-related stable disease (irSD), irPR, or irCR.
- Measure the time to new metastases (TTNM).

Any patient who has completed at least 3 cycles of study treatment are evaluable for this endpoint.

6.5 Exploratory Endpoints

Measure immune biomarkers (PDL-1) in CTCs (CellSearch™) and immune cells such as CAMLs (CellSieve™) and correlate with therapeutic benefit.

Any patient who has completed at least 3 cycles of study treatment are evaluable for this endpoint.

6.6 Statistical Analysis Plan

The primary hypothesis to be tested in this Phase 2 study is that the combination of Carboplatin AUC 5 intravenously and pembrolizumab (Keytruda®) will increase progression-free survival in patients with ≥ 5 CTCs compared with the historical median of 4.9 months [1]. The distribution of progression-free survival will be estimated using the Kaplan-Meier method. Median survival and an associated 95% confidence interval will be estimated from the KM analysis. The null hypothesis of no improvement in survival will be rejected if the lower bound of the confidence interval for median survival is greater than 4.9. All available data from all eligible treated patients will be included in efficacy analyses. Patients alive and not progressed at last follow-up will be censored at that time.

Change in CTCs longitudinally will be modeled using mixed effects regression. If possible, we will transform CTCs so that mixed effects models for normally distributed outcomes may be used. If a suitable transformation cannot be found or if there is zero-inflation, linear mixed effects models for zero-inflated outcomes as implemented in Mplus software will be used. Analysis of all time-to-event outcomes (OS, PFS, and TTNM) will begin with Kaplan-Meier plots and estimation of median survival times and associated 95% confidence intervals.

Association with baseline CTCs and other markers will be evaluated using log-rank tests with markers dichotomized at the median or other clinically relevant cut point. Analysis of association of baseline markers adjusted for other clinical characteristics will be performed using Cox proportional hazards regression or accelerated failure-time models, as appropriate. Correlations among markers for secondary endpoints will be estimated using Pearson or Spearman correlation coefficients.

Prior to Amendment 2 (dated July 30, 2018), the first six patients who were treated with sequential pembrolizumab followed by combination therapy and included endocrine-refractory disease will be used only to assess safety. This has been completed and there were no dose limiting toxicities in these patients. As the protocol is written, 38 patients will be accrued in order to obtain 32 patients who have been treated and are evaluable for progression-free survival (PFS). These numbers will be applied to the amended patient population. The null hypothesis will still be rejected if there is evidence that the median PFS exceeds 4.9 months.

6.7 Definitions

Response to treatment will be assessed using RECIST guidelines. Response and progression will be assessed in this study via physical exam and imaging. Lesions are defined as either measurable or non-measurable using the criteria provided below.

6.7.1 Lesion Measurement

Measurable Lesions - Must be accurately measured in at least one dimension (greatest diameter) with a minimum size of 10mm by CT scan (cuts of 5mm or less), MRI, or physical exam (ideally using calipers). Bone only disease is allowed, however, must have an osteoblastic component that can be easily measured.

Non-measurable Lesions - Defined as all other lesions less than 10mm. Examples of non-measurable lesions include leptomeningeal disease, ascities, pleural/pericardial effusions, inflammatory breast disease, and lymphangitic involvement of skin or lung.

6.7.2 Response criteria

Complete Response - Disappearance of all lesions.

Partial Response - At least a 30% decrease in the sum of the diameters of target lesions, taking as reference the baseline sum diameters.

Progressive Disease - At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression.)

Stable Disease - Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum of diameters while on study.

6.7.3 *Immune-related RECIST*

In addition to above, tumor response will be assessed by the irRECIST for investigational purposes, not assessment of primary objective [22]. Tumor response by irRECIST is defined as an immune-related PR (irPR) or CR (irCR) over a period of at least 4 weeks. We will define immune-related clinical benefit rate as immune-related stable disease (irSD), irPR, or irCR).

irRECIST is RECIST 1.1 adapted as described below to account for the unique tumor response seen with immuno-therapeutics. irRECIST will be used by site investigators and local radiology review to assess tumor response and progression, and make treatment decisions. This data will be collected in the clinical database.

irRECIST will be used by the central imaging vendor, however, this evaluation will be done retrospectively. irRECIST takes into account the clinical condition/stability of subjects, as described in the Table 7 below, in addition to response or progression via tumor imaging.

Clinically stable is defined by the following criteria:

- Absence of symptoms and signs indicating clinically significant progression of disease, including worsening of laboratory values
- No decline in ECOG performance status
- Absence of rapid progression of disease
- Absence of progressive tumor at critical anatomical sites (e.g., cord compression) requiring urgent alternative medical intervention

Table 7: irRECIST: Tumor Imaging and Treatment after 1st Radiologic Evidence of PD or SD, CR or PR

	Clinically Stable		Clinically Unstable	
	Imaging	Treatment	Imaging	Treatment
1 st radiologic evidence of PD	Repeat imaging at $\geq$ 4 weeks at site to confirm PD	May continue study treatment at the Investigator's discretion while awaiting confirmatory scan by site	Repeat imaging at $\geq$ 4 weeks to confirm PD per physician discretion only	Discontinue treatment
Repeat scan confirms PD	No additional imaging required	Discontinue treatment (exception is possible upon consultation with Sponsor).	No additional imaging required	N/A
Repeat scan shows SD, PR or CR	Continue regularly scheduled imaging assessments	Continue study treatment at the Investigator's discretion. Next	Continue regularly scheduled imaging assessments	May restart study treatment if condition has improved and/or clinically stable per Investigator's discretion.

		scan should occur according to the every 9 week (63 ± 7 days) imaging schedule.		Next scan should occur according to the every 9 week (63 ± 7 days) imaging schedule.
<i>Note: Clinical stability / instability will be a clinical decision and this should be documented in the patient's chart with proper supporting documents.</i>				

7 DRUG INFORMATION

7.1 Pembrolizumab

The investigator shall take responsibility for and shall take all steps to maintain appropriate records and ensure appropriate supply, storage, handling, distribution and usage of investigational product in accordance with the protocol and any applicable laws and regulations.

7.1.1 Other names

Keytruda™; MK-3475

7.1.2 Classification – type of agent

Humanized X PD-1_mAb (H409A11) IgG4

7.1.3 Mode of action

Highly selective anti-PD-1 humanized monoclonal antibody which inhibits programmed cell death-1 (PD-1) activity by binding to the PD-1 receptor on T-cells to block PD-1 ligands (PD-L1 and PD-L2) from binding. Blocking the PD-1 pathway inhibits the negative immune regulation caused by PD-1 receptor signaling (Hamid, 2013). Anti-PD-1 antibodies (including pembrolizumab) reverse T-cell suppression and induce antitumor responses.

7.1.4 Storage and stability

Please refer to pharmacy manual for detailed instructions

Clinical supplies must be stored in a secure, limited-access location under the storage conditions specified on the label.

Store Pembrolizumab vials under refrigeration at 2 to 8 degrees Celsius (°C) or 36 to 46 degrees Fahrenheit (°F). Do not freeze; do not shake.

Prior to reconstitution, the vials of lyophilized powder can be out of refrigeration (temperatures at or below 25°C (77°F)) for up to 24 hours.

Storage following reconstitution and dilution: Store at room temperature for a cumulative time up to 4 hours (this includes room temperature storage of the diluted infusion solution and the duration of infusion). IV bags and/or reconstituted vials can be also stored refrigerated at 2—8 degrees °C (36—46 degrees °F) for up to 20 hours. If refrigerated, allow the vials and/or IV bags to come to room temperature prior to use. Do not freeze.

Receipt and dispensing of trial medication must be recorded by an authorized person at the trial site.

Clinical supplies may not be used for any purpose other than that stated in the protocol

7.1.5 Protocol dose specifics

200mg IV on Day 1 of a 21 day cycle until disease progression or unacceptable toxicity. Pembrolizumab 200 mg will be administered as a 30 minute IV infusion every 21 days. Sites should make every effort to target infusion timing to be as close to 30 minutes as

possible. However, given the variability of infusion pumps from site to site, a window of -5 minutes and +10 minutes is permitted (i.e., infusion time is 30 minutes: -5 min/+10 min).

7.1.6 *Preparation*

Please refer to pharmacy manual for detailed instructions.

Pembrolizumab is provided as a white to off-white lyophilized powder (50 mg/vial) or as a liquid solution (100 mg/vial) in Type I glass vials intended for single use only.

Pembrolizumab is formulated with L-histidine as buffering agent, polysorbate 80 as surfactant, sucrose as stabilizer/tonicity modifier, and hydrochloric acid (HCl) and/or sodium hydroxide (NaOH) for pH adjustment (if necessary).

Pembrolizumab Powder for Solution for Infusion is reconstituted with 2.3 mL sterile water for injection (SWFI) to yield a 2.4 mL solution containing 25 mg/mL of pembrolizumab prior to further dilution. Pembrolizumab vial contains an excess fill of 10 mg (equivalent to 0.4 mL of reconstituted solution) to ensure the recovery of label claim of 50 mg/2 mL pembrolizumab per vial.

The product after **reconstitution with sterile water for injection** and the liquid drug product is a clear to opalescent solution, which may contain extraneous and proteinaceous particulates. The reconstituted drug product solution and liquid drug product are to be further diluted with normal saline in polyvinyl chloride (PVC) or non-PVC IV bags to achieve final concentration between 1 mg/mL and 10 mg/mL. Reconstituted vials and infusion solutions should be immediately administered after preparation. If not used immediately, reconstituted vials and/or IV bags may be stored at 2-8 °C for up to a cumulative time of 20 hours. If refrigerated, the vials and/or IV bags should be allowed to equilibrate to room temperature prior to subsequent use. Pembrolizumab solutions may be stored at room temperature for a cumulative time of up to 4 hours. This includes room temperature storage of reconstituted drug product solution and liquid drug product in vials, storage of infusion solution in the IV bag and the duration of infusion.

7.1.7 *Route of administration for this study*

Intravenous (IV)

7.1.8 *Incompatibilities*

None known

Refer to pharmacy manual for list of infusion set materials compatible with pembrolizumab.

7.1.9 *Availability and supply*

Merck will supply pembrolizumab (investigational / clinical supply) directly to the NU Investigational Pharmacy at no charge to subjects participating in this clinical trial. Drug orders are requested using the Merck approved pembrolizumab order form and sent to the centralized email ordering mailbox at onciisdrugorder@merck.com with a copy to the current Merck in-house contact for the study.

The treating investigator shall take responsibility for and shall take all steps to maintain appropriate records and ensure appropriate supply, storage, handling, distribution, and usage of investigational product in accordance with the protocol and any applicable laws and regulations.

Product Name & Potency	Dosage Form
Pembrolizumab 50 mg	Lyophilized Powder for Injection
Pembrolizumab 100 mg/ 4mL	Solution for Injection

7.1.10 *Restrictions during pregnancy and lactation*

Based on its mechanism of action, pembrolizumab can cause fetal harm when administered to a pregnant woman.

If a subject inadvertently becomes pregnant while on treatment with pembrolizumab, the subject will immediately be removed from the study. The site will contact the subject at least monthly and document the subject's status until the pregnancy has been completed or terminated. The outcome of the pregnancy will be reported to the Sponsor and to Merck without delay and within 24 hours to the Northwestern University and within 2 working days to Merck if the outcome is a serious adverse experience (e.g., death, abortion, congenital anomaly, or other disabling or life-threatening complication to the mother or newborn).

The study investigator will make every effort to obtain permission to follow the outcome of the pregnancy and report the condition of the fetus or newborn to Northwestern University. If a male subject impregnates his female partner the study personnel at the site must be informed immediately and the pregnancy reported to the Northwestern University and to Merck and followed as described above and in Section 7.0.

It is unknown whether pembrolizumab is excreted in human milk. Since many drugs are excreted in human milk, and because of the potential for serious adverse reactions in the nursing infant, subjects who are breast-feeding are not eligible for enrollment.

7.1.11 *Side effects*

Pembrolizumab is generally well tolerated and demonstrates a favorable safety profile in comparison to chemotherapy. Pembrolizumab is an immunomodulatory agent, and based on this mechanism of action, immune mediated adverse events are of primary concern. Important identified risks for pembrolizumab are of an immune mediate nature, including: pneumonitis, colitis, thyroid disorders (hypothyroidism/hyperthyroidism), hepatitis, hypophysitis, Type I diabetes mellitus, uveitis, and nephritis. After a recent review of data, events newly characterized as identified risks also include pancreatitis, myositis, and severe skin reaction; these are included in the reference safety information in the current IB. The majority of immune-mediated adverse events were mild to moderate in severity, were manageable with appropriate care, and rarely required discontinuation of therapy.

In addition to the previously noted identified risks, infusion-related reactions are a risk but are not considered immune mediated; these are also further described in the current IB.

Most common (seen in >10% of patients):

- Fatigue
- Headache
- Chills
- Difficulty sleeping
- Dizziness
- Leg swelling
- Itching and/or rash
- Elevated blood sugar level
- High triglycerides (type of cholesterol)
- Lower levels of calcium or sodium in the blood
- Nausea and/or vomiting
- Decreased appetite
- Diarrhea or constipation
- Pain in the abdomen
- Cough
- Shortness of breath

- Fever

Less common (seen in 1 to 10% of patients):

- Skin infection
- Abnormal thyroid function (decreased or increased)
- Colitis (inflammation or infection of the colon [large intestine])
- Sepsis (infection in the bloodstream)
- Kidney failure
- Pneumonitis (inflammation of the lung) or pneumonia (infection in the lung)

Rare but serious or life-threatening (seen in <1% of patients)

- *Immune system attacking adrenal gland (adrenocortical insufficiency), joints (arthritis), liver (hepatitis), kidneys (nephritis), skin (dermatitis) muscles (myositis), eyes (optic neuritis or uveitis), pancreas (pancreatitis), brain (partial epilepsy), or other organs causing possibly irreversible damage to these organs*

Please refer to the current version of the Investigator's Brochure for a complete list of adverse events.

7.1.12 *Nursing implications*

Pembrolizumab will be administered first, followed sequentially by carboplatin. (Refer to section 4.2 for details)

Pembrolizumab will be administered as an IV infusion ((section 8.1.6) over approximately 30 minutes. Every effort should be made to target infusion timing to be as close to 30 minutes as possible. However, a window of -5 minutes and +10 minutes is permitted (i.e., infusion time is 30 minutes: -5 min/+10 min). Sterile, non-pyrogenic, low-protein binding 0.2 to 5 µm in-line filter made of polyethersulfone (PES) must be used during administration.

When an IV bag is used for the infusion, the IV line will be flushed with a volume of normal saline equal to the priming volume of the infusion set used after the contents of the IV bag are fully administered

Since the compatibility of pembrolizumab with other IV medications and solutions, other than normal saline (0.9% [w/v] sodium chloride for injection) and D5W, is not known, the Pembrolizumab solution should not be infused through an IV line in which other solutions or medications are being administered. The date, start time, interruption, and completion time of Pembrolizumab administration must be recorded in the source documents. Any unused solution for injection should not be used for another infusion of the same or different subject.

7.1.13 *Return and retention of study drug*

The investigator is responsible for keeping accurate records of the clinical supplies received from Merck or designee, the amount dispensed to and returned by the subjects and the amount remaining at the conclusion of the trial.

Upon completion or termination of the study, all unused pembrolizumab vial(s) shall be returned to the designated facility for destruction.

Please refer to pharmacy manual for the address of the designated facility.

All used and/or partially used investigational product will be destroyed at the site per institutional policy. It is the Investigator's responsibility to arrange for disposal of all empty containers, provided that procedures for proper disposal have been established according to applicable federal, state, local and institutional guidelines and procedures, and provided that appropriate records of disposal are kept.

7.2 Carboplatin

7.2.1 *Other names*

CBDCA, Paraplatin, JM-8, NSC-241240

7.2.2 *Classification – type of agent*

Second-generation tetravalent organic platinum compound

7.2.3 *Mode of Action*

Like cisplatin, carboplatin binds to DNA, thereby inhibiting DNA synthesis, in a cell cycle nonspecific manner. Carboplatin must first undergo activation to produce antineoplastic activity. Bidentate carboxylate ligands of carboplatin are displaced by water forming (aquation) positively charged platinum complexes which bind to nucleophilic sites in DNA, such as the O-6 position on guanine. Carboplatin produces predominantly interstrand DNA crosslinks rather than DNA-protein crosslinks. Intrastrand crosslinks result from the formation of adducts between the activated platinum complexes of the drug and the N-7 atom (not exclusively) atom on guanine to produce 1, 2 intrastrand links between adjacent guanine molecules, between neighboring guanine and adenosine molecules, or between neighboring guanine molecules. Interstrand cross-linking within the DNA helix also occurs. Platinum adducts may inhibit DNA replication, transcription and ultimately cell division.

7.2.4 *Storage and stability*

Intact vials are stored at room temperature protected from light. The reconstituted solution is stable for at least 24 hours. When further diluted in glass or polyvinyl plastic to a concentration of 10mg/mL with normal saline or 5% dextrose carboplatin is stable for 8 hours at 25 degrees C. Stability with further dilution to 0.5mg/mL has been reported for up to 8 hours. Other stability data indicate that carboplatin is stable for up to 24 hours and may be refrigerated, however, the manufacturer recommends that reconstituted solutions be discarded after 8 hours due to the lack of preservative in drug formulation.

7.2.5 *Protocol dose specifics*

Carboplatin will be given by IV at an area under the curve (AUC) dose of 5. Routine premedication should include at least a 5 – HT3 antagonist and dexamethasone. Carboplatin dose calculation is be per institutional standard and based on the serum creatinine of the day of visit.

7.2.6 *Preparation*

Add 5, 15, or 45 mL sterile water, normal saline, or 5% dextrose to the 50, 150, or 450 mg vial, respectively. The resulting solution contains 10 mg/mL. The desired dose is further diluted, usually in 5% dextrose.

7.2.7 *Route of administration for this study*

IV over 30 minutes.

7.2.8 *Incompatibilities*

Aluminum displaces platinum from the carboplatin molecule, resulting in the formation of a black precipitate and loss of potency. Carboplatin solutions should not be prepared or administered with needles, syringes, catheters, or IV administration sets containing aluminum parts that might be in contact with the drug.

7.2.9 *Availability & Supply*

Commercially available.

Concomitant myelosuppressive drugs or radiation therapy may potentiate the hematologic toxicity of carboplatin. Concomitant nephrotoxic drugs may potentiate the nephrotoxicity of carboplatin, particularly when carboplatin is given in high-dose chemotherapy regimens.

7.2.10 Side effects

1. Hematologic: Thrombocytopenia (dose limiting), neutropenia, leukopenia, anemia.
2. GI: Nausea and vomiting (frequent but less severe than with cisplatin), treatable with appropriate antiemetic prophylaxis.
Anorexia, diarrhea and constipation have also been reported.
3. Dermatologic: Rash, urticaria. Rarer reactions include alopecia, mucositis, and hypersensitivity reactions.
4. Hepatic: Abnormal liver function tests, usually reversible with standard doses.
5. Neurologic: Rarely peripheral neuropathy is seen. May be more common in patients greater than 65 years of age. May also be cumulative, especially in patients with prior cisplatin treatment.
Ototoxicity (rare).
6. Renal: Elevations in serum creatinine, BUN; electrolyte loss (Mg, K, Na, Ca).
7. Miscellaneous: Pain, asthenia, flu-like syndrome.

7.2.11 Nursing implications

1. Monitor CBC and platelet count routinely.
2. Premedicate with antiemetics – prophylaxis with a 5HT3 receptor antagonist and dexamethasone (+/- aprepitant) is standard.
3. Monitor fluid status – maintain adequate hydration.
4. Assess skin/mucous membranes.
5. Assess for signs of peripheral neuropathy – coordination, sensory and hearing loss.

8 ADVERSE EVENTS

This study will be conducted in compliance with the Data Safety Monitoring Plan ([DSMP](#)) of the Robert H. Lurie Comprehensive Cancer Center of Northwestern University (please refer to Appendices for additional information). The level of risk attributed to this study requires high intensity monitoring, as outlined in the [DSMP](#). In addition, the study will abide by all safety reporting regulations, as set forth in the Code of Federal Regulations.

8.1 Adverse Event Monitoring

AE data collection and reporting, which are required as part of every clinical trial, are done to ensure the safety of subjects enrolled in the studies as well as those who will enroll in future studies using similar agents. AEs are reported in a routine manner at scheduled times during a trial (see Section 5 for time points). In addition, certain AEs must be reported in an expedited manner to allow for optimal monitoring and patient safety and care.

All patients experiencing an AE, regardless of its relationship to study drug, will be followed until:

- the AE resolves or the symptoms or signs that constitute the AE return to baseline;
- any abnormal laboratory values have returned to baseline;
- there is a satisfactory explanation other than the study drug for the changes observed; or
- death.

8.2 Definitions & Descriptions**8.2.1 Adverse Event**

An adverse event (AE) is any untoward medical occurrence in a patient receiving study treatment and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of an experimental intervention, whether or not related to the intervention.

Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a preexisting condition that is temporally associated with the use of the Merck's product, is also an AE.

Changes resulting from normal growth and development that do not vary significantly in frequency or severity from expected levels are not to be considered AEs. Examples of this may include, but are not limited to, teething, typical crying in infants and children and onset of menses or menopause occurring at a physiologically appropriate time. Merck product includes any pharmaceutical product, biological product, device, diagnostic agent or protocol-specified procedure, whether investigational (including placebo or active comparator medication) or marketed, manufactured by, licensed by, provided by or distributed by Merck for human use.

AEs may occur during the course of the use of Merck product in clinical trials, or as prescribed in clinical practice, from overdose (whether accidental or intentional), from abuse and from withdrawal.

Progression of the cancer under study is not considered an AE. If patient experiences PD, they will come off study treatment. If patient experiences a SAE from drug toxicity follow toxicity management guidelines and/or physician discretion.

All AEs that occur after the consent form is signed but before treatment allocation/randomization must be reported by the investigator if they cause the subject to be excluded from the trial, or are the result of a protocol-specified intervention, including but not limited to washout or discontinuation of usual therapy, diet, placebo treatment or a procedure.

From the time of treatment allocation/randomization through 30 days following cessation of treatment, all AEs must be reported by the investigator. Such events will be recorded at each examination on the AEs eCRFs. The reporting timeframe for AEs meeting any serious criteria is described in section 8.3. The investigator will make every attempt to follow all subjects with non-SAEs for outcome.

Recording of AEs should be done in a concise manner using standard, acceptable medical terms. In general, AEs are not procedures or measurements, but should reflect the reason for the procedure or the diagnosis based on the abnormal measurement. Preexisting conditions that worsen in severity or frequency during the study should also be recorded (a preexisting condition that does not worsen is not an AE). Further, a procedure or surgery is not an AE; rather, the event leading to the procedure or surgery is considered an AE.

If a specific medical diagnosis has been made, that diagnosis or syndrome should be recorded as the AE whenever possible. However, a complete description of the signs, symptoms and investigations which led to the diagnosis should be provided. For example, if clinically significant elevations of liver function tests are known to be secondary to hepatitis, "hepatitis" and not "elevated liver function tests" should be recorded. If the cause is not known, the abnormal test or finding should be recorded as an AE, using appropriate medical terminology (e/g/ thrombocytopenia, peripheral edema, QT prolongation).

8.2.2 *Severity of AEs*

All non-hematologic AEs will be graded according to the NCI Common Terminology Criteria for Adverse Events (CTCAE) version 4.03. The CTCAE v4 is available at <http://ctep.cancer.gov/reporting/ctc.html>

If no CTCAE grading is available, the severity of an AE is graded as follows:

- Mild (grade 1): the event causes discomfort without disruption of normal daily activities.
- Moderate (grade 2): the event causes discomfort that affects normal daily activities.
- Severe (grade 3): the event makes the patient unable to perform normal daily activities or significantly affects his/her clinical status.
- Life-threatening (grade 4): the patient was at risk of death at the time of the event.
- Fatal (grade 5): the event caused death.

8.2.3 *Serious Adverse Events (SAEs)*

All SAEs, regardless of attribution, occurring from time of signed informed consent, through 90 days after the last administration of study drug, must be reported upon discovery or occurrence.

An SAE is defined in regulatory terminology as any untoward medical occurrence that:

- Results in *death*.
- If death results from (progression of) the disease, the disease should be reported as event (SAE) itself.
- Is life-threatening.
- The patient was at risk of death at the time of the event; it does not refer to an event that hypothetically might have caused death if it were more severe.
- Requires in-patient hospitalization or prolongation of existing hospitalization for $\geq$ 24 hours.
- Results in persistent or significant disability or incapacity.
- Is a congenital anomaly/birth defect.
- Is an important medical event.
- Any event that does not meet the above criteria, but that in the judgment of the investigator jeopardizes the patient, may be considered for reporting as a SAEs. The event may require medical or surgical intervention to prevent one of the outcomes listed in the definition of "Serious Adverse Event".
- For example: allergic bronchospasm requiring intensive treatment in an emergency room or at home; convulsions that may not result in hospitalization; development of drug abuse or drug dependency.
- **Note:** In addition to the above criteria, AEs meeting either of the below criteria, although not serious per ICH definition, are reportable to the Merck in the same timeframe as SAEs to meet certain local requirements. Therefore, these events are considered serious by Merck for collection purposes.
 - Is a new cancer (that is not a condition of the study);
 - Is associated with an overdose.

8.2.4 *Unanticipated Problems Involving Risks to Subject or Others*

A UPIRSO is a type of SAE that includes events that meet ALL of the following criteria:

- Is *unanticipated* in terms of nature, severity, or frequency
- Places the research subject or others at a different or *greater risk of harm*
- Is deemed to be *at least possibly related* to participation in the study.

8.3 Adverse Event Reporting

8.3.1 *Routine Reporting*

All routine AEs, such as those that are expected, or are unlikely or definitely not related to study participation, are to be reported on the appropriate eCRF. Routine AEs will be

reviewed by the Data Monitoring Committee (DMC) according to the study's phase and risk level, as outlined in the [DSMP](#).

8.3.2 *Determining if Expedited Reporting is Required*

This includes all events that occur within 30 days of the last dose of protocol treatment. Any event that occurs more than 30 days after the last dose of treatment and is attributed (possibly, probably, or definitely) to the agent(s) must also be reported accordingly.

- 1) Identify the type AE using the NCI CTCAE v 4.03.
- 2) Grade the AE using the NCI CTCAE v 4.03.
- 3) Determine whether the AE is related to the protocol therapy.
Attribution categories are as follows:
 - Definite: AE is clearly related to the study treatment.
 - Probable: AE is likely related to the study treatment.
 - Possible: AE may be related to the study treatment.
 - Unlikely: AE not likely to be related to the study treatment.
 - Unrelated: AE is clearly NOT related to the study treatment.
- 4) Determine the prior experience of the AE.
Expected events are those that have been previously identified as resulting from administration of the agent. An AE is considered unexpected, for expedited reporting purposes only, when either the type of event or the severity of the event is not listed in:
 - the current protocol
 - the drug package insert
 - the current Investigator's Brochure

8.3.3 *Expedited Reporting to the Northwestern University QAM/DMC*

All SAEs must be reported to the assigned QAM within 24 hours of becoming aware of the event. Completion of the NU CRO SAE Form, provided as a separate document, is required.

The completed form should assess whether or not the event qualifies as a UPIRSO. The report should also include:

- Protocol description and number(s)
- The patient's identification number
- A description of the event, severity, treatment, and outcome (if known)
- Supportive laboratory results and diagnostics
- The hospital discharge summary (if available/applicable)

All SAEs will be reported to, and reviewed by, the DMC at their next meeting.

8.3.4 *Expedited Reporting to the Northwestern University IRB*

- Any death of an NU subject that is unanticipated in nature and at least possibly related to study participation will be promptly reported to the NU IRB within 24 hours of notification.
- Any death of an NU subject that is actively on study treatment (regardless of whether or not the event is possibly related to study treatment)
- Any death of a non-NU subject that is unanticipated and at least possibly related and any other UPIRSOs will be reported to the NU IRB within 5 working days of notification.
- All other deaths of NU subjects not previously reported, other non-NU subject deaths that were unanticipated and unrelated, and any other SAEs that were not previously reported as UPIRSOs will be reported to the NU IRB at the time of annual continuing review.

8.3.5 *Reporting to the FDA*

The FDA will be notified within 7 calendar days of any SAE that is associated with study treatment, is unexpected, and is fatal or life-threatening.

The FDA will be notified within 15 calendar days of any SAE that is associated with the study treatment, unexpected, and serious but *not fatal or life-threatening*. This includes any previous SAEs that were not initially deemed reportable, but are later determined to meet the criteria for reporting (i.e. by the DSMC).

All other SAEs will be reported on an annual basis as part of the annual FDA report.

8.3.6 *Expedited Reporting to Merck*

For the time period beginning when the consent form is signed until treatment allocation/randomization, any SAE, or follow up to a SAE, including death due to any cause other than progression of the cancer under study that occurs to any subject must be reported within 24 hours to the NU QA and/or IRB (see section 7.3.3.1 and 7.3.3.2) and within 2 working days to Merck Global Safety if it causes the subject to be excluded from the trial, or is the result of a protocol-specified intervention, including but not limited to washout or discontinuation of usual therapy, diet, placebo treatment or a procedure. The NU CRO SAE form will be used.

For the time period beginning at treatment allocation/randomization through 90 days following cessation of treatment, or 30 days following cessation of treatment if the subject initiates new anticancer therapy, whichever is earlier, any SAE, or follow up to a SAE, including death due to any cause other than progression of the cancer under study whether or not related to the Merck product, must be reported within 24 hours to the S and within 2 working days to Merck Global Safety. The NU CRO SAE form will be used.

Additionally, any SAE, considered by an investigator who is a qualified physician to be related to Merck product that is brought to the attention of the investigator at any time following consent through the end of the specified safety follow-up period specified in the paragraph above, or at any time outside of the time period specified in the previous paragraph also must be reported immediately to the PI and to Merck Global Safety.

All subjects with SAEs must be followed up for outcome.

SAE reports and any other relevant safety information are to be forwarded to the Merck Global Safety facsimile number: +1-215-993-1220.

A copy of all 15 Day Reports and Annual Progress Reports is submitted as required by FDA, European Union (EU), Pharmaceutical and Medical Devices agency (PMDA) or other local regulators. Investigators will cross reference this submission according to local regulations to the Merck Investigational Compound Number (IND, CSA, etc.) at the time of submission. Additionally investigators will submit a copy of these reports to Merck & Co., Inc. (Attn: Worldwide Product Safety; FAX 215 993-1220) at the time of submission to FDA (see section 8.3.5).

8.4 Other Events Requiring Expediting Reporting

8.4.1 *Overdose*

For purposes of reporting to Merck, an overdose of pembrolizumab will be defined as any dose of 1,000 mg or greater (≥ 5 times the indicated dose). NOTE: the Northwestern University IRB requires reporting of any overdose according to their Reportable New Information criteria. No specific information is available on the treatment of overdose of pembrolizumab. In the event of overdose, the subject should be observed closely for

signs of toxicity. Appropriate supportive treatment should be provided if clinically indicated.

If an AE is associated with (“results from”) the overdose of a Merck product, the AE is reported as a SAE, even if no other seriousness criteria are met.

If a dose of Merck’s product meeting the protocol definition of overdose is taken without any associated clinical symptoms or abnormal laboratory results, the overdose is reported as a non-serious ECI, using the terminology “accidental or intentional overdose without adverse effect.”

All reports of overdose with and without an AE must be reported using the NU CRO SAE form within 24 hours to the PI and within 2 working days hours to Merck Global Safety. (Attn: Worldwide Product Safety; FAX 215 993-1220).

(Note: This does not apply to Northwestern University, because, as per NU IRB regulations, any amount of overdose will be reported as a Reportable New Information [RNI])(Refer Section 10.6.2).

8.4.2 *Reporting of Pregnancy and Lactation*

Although pregnancy and lactation are not considered AE, it is the responsibility of investigators or their designees to report any pregnancy or lactation in a subject (spontaneously reported to them) that occurs during the trial.

Pregnancies and lactations that occur after the consent form is signed but before treatment allocation/randomization must be reported by the investigator if they cause the subject to be excluded from the trial, or are the result of a protocol-specified intervention, including but not limited to washout or discontinuation of usual therapy, diet, placebo treatment or a procedure.

Pregnancies and lactations that occur from the time of treatment allocation/randomization through 120 days following cessation of treatment, or 30 days following cessation of treatment if the subject initiates new anticancer therapy, whichever is earlier, must be reported by the investigator. All reported pregnancies must be followed to the completion/termination of the pregnancy. Pregnancy outcomes of spontaneous abortion, missed abortion, benign hydatidiform mole, blighted ovum, fetal death, intrauterine death, miscarriage and stillbirth must be reported as serious events (Important Medical Events). If the pregnancy continues to term, the outcome (health of infant) must also be reported.

Such events must be reported using the NU CRO SAE form within 24 hours to the PI and within 2 working days to Merck Global Safety. (Attn: Worldwide Product Safety; FAX 215 993-1220)

8.4.3 *Events of Clinical Interest*

Selected AEs and SAEs are also known as Events of Clinical Interest (ECI) and must be reported within 24 hours to the PI and within 2 working days to Merck Global Safety. (Attn: Worldwide Product Safety; FAX 215 993-1220).

For the time period beginning when the consent form is signed until treatment allocation/randomization, any ECI, or follow up to an ECI, that occurs to any subject must be reported within 24 hours to the PI and within 2 working days to Merck Global Safety if it causes the subject to be excluded from the trial, or is the result of a protocol-specified intervention, including but not limited to washout or discontinuation of usual therapy, diet, placebo treatment or a procedure.

For the time period beginning at treatment allocation/randomization through 90 days following cessation of treatment, or 30 days following cessation of treatment if the subject initiates new anticancer therapy, whichever is earlier, any ECI, or follow up to an ECI, whether or not related to Merck product, must be reported within 24 hours to the PI and within 24 hours to Merck Global Safety.

Events of clinical interest for this trial include:

1. an overdose of Merck product that is not associated with clinical symptoms or abnormal laboratory results.
2. an elevated AST or ALT lab value that is greater than or equal to 3X the upper limit of normal and an elevated total bilirubin lab value that is greater than or equal to 2X the upper limit of normal and, at the same time, an alkaline phosphatase lab value that is less than 2X the upper limit of normal, as determined by way of protocol-specified laboratory testing or unscheduled laboratory testing.*

*Note: The purpose of the criteria is to specify a threshold of abnormal hepatic tests that may require an additional evaluation for an underlying etiology.

9 CORRELATIVES/SPECIAL STUDIES

Blood and tissue will be collected as part of study procedures (see Section 5). Details on sample collection, processing, storage and shipment can be found in the Lab Manual.

9.1 Sample Collection Guidelines

Please refer to Lab Manual for details on sample collection.

9.2 Sample Processing, Storage, and Shipment

Please refer to Lab Manual for details on sample processing, storage and shipment.

9.3 Assay Methodology

9.3.1 CTCs enumeration and PDL-1 evaluation

CTC isolation and enumeration will be performed using the US Food and Drug Administration-approved CellSearch™ technology (Janssen Diagnostics). This technology consists of a semi-automated system for the preparation of a sample [35] and is used with the CellSearch™ Epithelial Cell Kit. The procedure enriches the sample for cells expressing the epithelial-cell adhesion molecule with antibody-coated magnetic beads, and labels the cells with the fluorescent nucleic acid dye 4, 2-diamidino-2-phenylindole dihydrochloride (DAPI). Fluorescently labeled monoclonal antibodies specific for leukocytes (CD45-allophycocyan) and epithelial cells (cytokeratin 8, 18, 19-phycoerythrin) are used to distinguish epithelial cells from leukocytes. A PDL-1 expression kit has been developed and is currently tested in clinical trial evaluating PDL-1 or PD-1 therapeutic targeting antibodies.

9.3.2 CTCs and CAMLs using CellSieve™

The CellSieve Microfiltration assay isolates CTCs and CAMLs by size exclusion, identifying CTCs or CAMLs by their morphological features and the phenotypic expression of CD14, cytokeratins 8, 18, and 19, CD45 and DAPI. The low-pressure vacuum system uses a specialized, holder attached to a syringe pump, to provide steady state flow of samples. Peripheral blood (7.5 mL) collected in CellSave preservative tubes is prefixed for 15 min, placed into the 30-mL syringe, and drawn through the filter in ~3 min. The filter is then washed, post-fixed, and permeabilized for 15 min. The filter and cells are stained with an antibody mixture and blocking buffer of FITC-conjugated anti-cytokeratin 8, 18, 19, phycoerythrin (PE)-conjugated CD14, and Cy5-conjugated CD45 (Creatv MicroTech, Inc.).

CTCs are CD45/CD14 negative cells with a pleomorphic nucleus and a filamentous cytokeratin positive phenotype. CAMLs are morphologically identified as single cells with an enlarged nuclear profile (14–64 µm in diameter) or separated polymorphic nuclei contained within the cell. In addition, the nuclei had to be surrounded by a larger CD14 signal (21–300 µm in length). CAMLs can be identified further by vacuoles containing DAPI+ and/or cytokeratin (Figure 2).

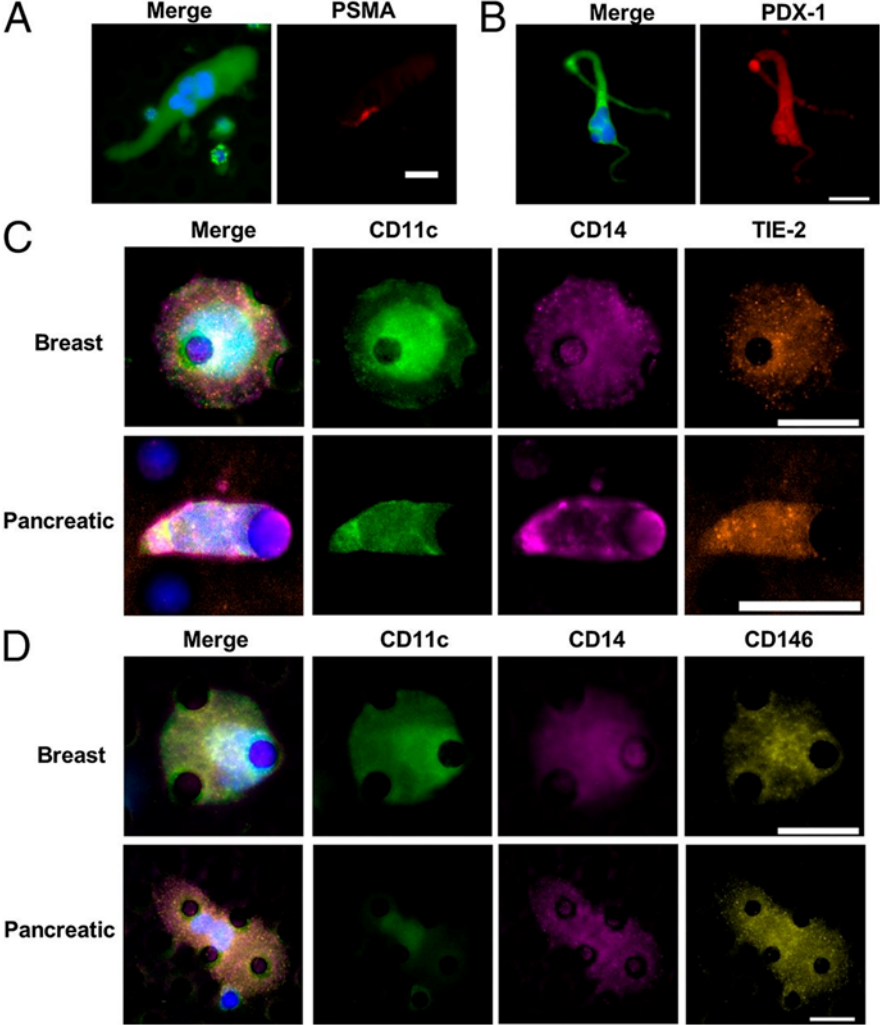


Figure 3.

In our most recent development and collaboration we have been able to perform fluorescently labeled stain of single CTCs for identification and monitoring of immune markers, including PD-L1 (Figure 3).

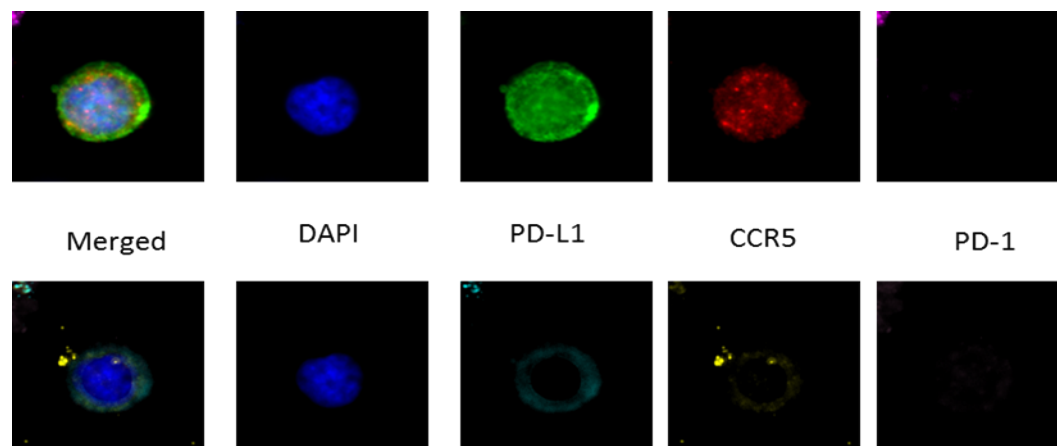


Figure 4.

9.3.3 Next Generation Sequencing for T-Cell Receptor clonality

The use of immune checkpoint inhibitors demonstrated clinical activity in several malignancies including melanoma, NSCLC and most recently breast and bladder cancers. There are recent laboratory data suggesting that use of anti-PD1,-PDL-1 and CTL4-inhibitors affect T-cell receptor clonality and diversity contributing to increase efficacy and sometimes toxicity. These effects can be measured in tissue as tumor-infiltrating lymphocytes (TILs) diversity and peripheral blood [36-38].

The evaluation of T-cell receptor clonality and their predictive value was assessed in tumor TILs using NGS (Adaptive Biotech). Patients with advanced melanoma treated with PDL-1 inhibitor, nivolumab demonstrated response to the investigational agent in patients whose tumor expressed PDL-1 but also had expanded TILs clone after treatment [37].

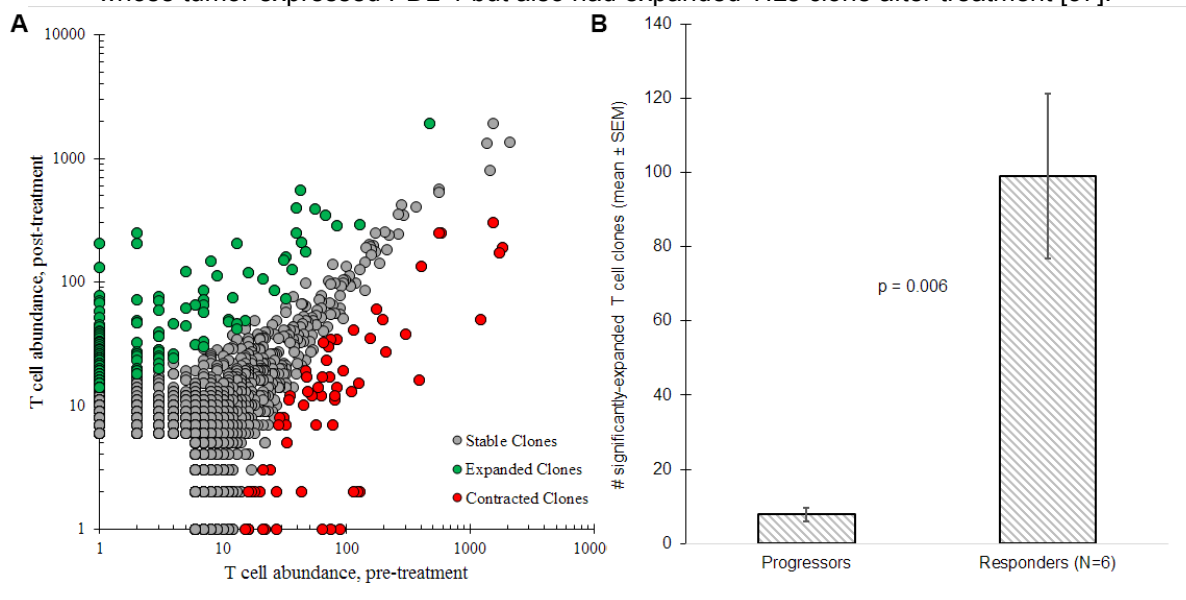


Figure 5.

Post-treatment clonality appears predictive of response and benefit after use of BRAF-inhibitors suggesting that targeted therapies may increase tumor immunogenicity [36]. These and other data seem to suggest the need to measure the dynamic changes occurring pre-treatment and after immune therapies as predictor of response. Moreover, the combinations with chemotherapy and targeted therapies may be associated with

change in mutation burden or other tumor-related effects contributing to increase immunogenicity. Those changes would be informative when using both tissue (when available) and peripheral blood evaluations.

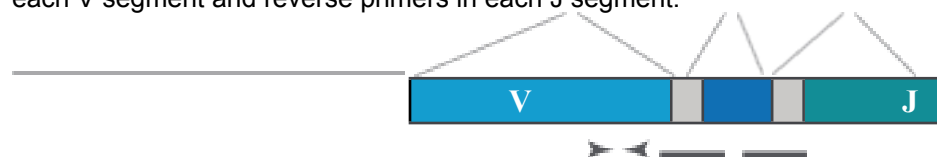
9.3.4 Human and Mouse T-cell Receptor Sequencing

The cellular adaptive immune system generates a remarkable breadth of diversity in antigen-specific T-cell receptors (TCR) by combinatorial recombination of gene segments in lymphocytes. The TCR is composed of two peptide chains, encoded by the T-cell receptor alpha and beta (TCRA, TCRB) or T-cell receptor gamma and delta (TCRG, TCRD) genes. There are thus two types of T-cell receptors, $\alpha\beta$ and $\gamma\delta$, that differ by the TCR heterodimer type and immune function. The antigenic specificity of T-lymphocytes is in large part determined by the amino acid sequence in the hypervariable complementarity-determining region 3 (CDR3) of T-cell receptors. The existence of multiple V and J gene segments at the T-cell loci permits a large combinatorial diversity in receptor composition; and the non-templated insertion or deletion of nucleotides at the V-J, V-D, and D-J junctions further adds to the potential diversity of receptors that can be encoded. Because of the potential diversity of receptors (a healthy adult has approximately 10 million different TCRB chains contained within their 10 [39, 40] circulating T-cells[39]) it is highly improbable to randomly converge on the same TCRB nucleotide CDR3 sequence, effectively making each CDR3 sequence a unique tag for a T-cell clone.

Adaptive Biotechnologies' immunoSEQ Assay, a multiplex PCR-based method that amplifies rearranged TCR CDR3 sequences (Fig. 6) and exploits the capacity of high throughput sequencing technology, characterizes tens of thousands of TCRB CDR3 chains simultaneously. The primary obstacle to addressing amplification bias in repertoire analysis has been the lack of any gold standard in which the exact concentration of each target is known prior to amplification. To address this problem, Adaptive Biotechnologies developed a rigorous PCR amplification bias-control process, ensuring a quantitative read-out of the adaptive immune repertoire². Additionally, because the technology utilizes genomic DNA, the frequency of sequenced CDR3 chains is highly representative of the relative frequency of each CDR3 sequence in the sample population of T-cells. Thus, the assay captures the full TCR repertoire including specific individual clones. The immunoSEQ Assay provides a novel method to identify and track the presence and frequency of common and rare clones in the context of the total adaptive immune system.



Figure 6: IGH sequencing assay. (A) Multiple V, D and J segments exist in the germline genome, non-templated diversity is introduced at the junctions by insertion of random nucleotides (shown as N). (B) The assay uses a multiplex PCR with forward primers in each V segment and reverse primers in each J segment.



10 STUDY MANAGEMENT

10.1 Institutional Review Board (IRB) Approval and Consent

It is expected that the IRB will have the proper representation and function in accordance with federally mandated regulations. The IRB should approve the consent form and protocol.

In obtaining and documenting informed consent, the investigator should comply with the applicable regulatory requirement(s), and should adhere to Good Clinical Practice (GCP) and to ethical principles that have their origin in the Declaration of Helsinki.

Before recruitment and enrollment onto this study, the patient will be given a full explanation of the study and will be given the opportunity to review the consent form. Each consent form must include all the relevant elements currently required by the FDA Regulations and local or state regulations. Once this essential information has been provided to the patient and the investigator is assured that the patient understands the implications of participating in the study, the patient will be asked to give consent to participate in the study by signing an IRB approved consent form.

Prior to a patient's participation in the trial, the written informed consent form should be signed and personally dated by the patient and by the person who conducted the informed consent discussion.

10.2 Amendments

Amendments to the protocol will be initiated and maintained by the assigned Medical Writer. Requests for revisions may come from multiple sources, including but not limited to the Principal Investigator, study team, drug company, or FDA. All amendments will be subject to the review and approval of the appropriate local, institutional, and governmental regulatory bodies, as well as by Merck Sharp and Dohme Corp. Amendments will be distributed by the lead institution (Northwestern) to all participating sites upon approval by the Northwestern University IRB.

10.3 Registration Procedures

For potential patients of this study, study teams are asked to inform the QAM of the date and time that the patient will need to be registered (croqualityassurance@northwestern.edu).

BEFORE a patient can be treated on study, the following items must be completed and submitted to confirm eligibility and receive a subject identification number:

- Patient's signed and dated informed consent form (upload to NOTIS and keep original hard copy in a secure location/study chart)
- Copy of the pathology report (upload to NOTIS)
- Signed and dated Eligibility Checklist (upload to NOTIS and keep original hard copy in a secure location/study chart)

The assigned QAM will review the registration documents. Once review is complete, he or she will register the patient, assign a subject identification number, provide a cohort assignment (as applicable) and send a confirmation of registration to involved personnel. Registration will then be complete and the patient may begin study treatment

10.4 Data Submission

Data collection for this study will be done through [NOTIS](#). Access to the trial in NOTIS is granted to appropriate roles identified at the time of participating site activation, or upon request. Site users will not be able to access the study in NOTIS until all required and study specific trainings are completed.

Once a patient is confirmed and registered to the study, eCRFs should be submitted according to the study procedures table. Generally, for all phase II patients, data are due with 10 days of a visit or end of cycle. A set amount of data may also be requested for any screen failures, as is

defined by the study. In most instances, this will include collection of adverse events and baseline data from the time of registration to the date of screen failure.

10.5 Instructions for Participating Site

Before the study can be initiated at any site, the following documentation must be provided to the Clinical Trials Office at Northwestern University (as applicable):

- Completed feasibility assessment(s) to verify site's capacity to support a Northwestern sponsored trial
- Signed copy of Northwestern University's Data Participating Site Acknowledgement which details data submission guidelines
- Draft consent form for review and approval prior to submission to the local IRB
- A copy of the official IRB approval letter for the protocol and informed consent
 - A copy of the IRB approved informed consent
- Pertinent credentials (CVs, MLs, CITI & GCP Training and FDFs) for the local PI and any sub-investigators who will be involved in the study at the site
- Form FDA 1572 appropriately filled out and signed with appropriate supporting certifications

Additional activities may be required prior to site activation (i.e. contract execution, study-specific training, and delegation of authority log). Full requirements will be outlined in the study start-up packet upon successful completion of a feasibility assessment.

10.6 Data Management and Monitoring/Auditing

This study will be conducted in compliance with the Data Safety Monitoring Plan ([DSMP](#)) of the Robert H. Lurie Comprehensive Cancer Center of Northwestern University (please refer to NOTIS for additional information). The level of risk attributed to this study requires high intensity monitoring, as outlined in the [DSMP](#). The assigned QAM, with oversight from the Data Monitoring Committee, will monitor this study in accordance with the study phase and risk level. In addition, the study will abide by all safety reporting regulations, as set forth in the Code of Federal Regulations.

10.7 Adherence to the Protocol

Except for an emergency situation in which proper care for the protection, safety, and well-being of the study patient requires alternative treatment, the study shall be conducted exactly as described in the approved protocol.

10.7.1 *Emergency Modifications*

Investigators may implement a deviation from, or a change of, the protocol to eliminate an immediate hazard(s) to trial subjects without prior IRB approval.

For any such emergency modification implemented, an IRB modification form must be completed within 5 business days of making the change, and the QAM must be notified within 24 hours of such change. Such modifications also need to be reported to the FDA, as applicable, within the appropriate timelines.

10.7.2 *Other Protocol Deviations*

A protocol deviation is any change, divergence, or departure from the study design or procedures of a research protocol that is under the investigator's control and that has not been approved by the Institutional Review Board (IRB). Protocol deviations must be reported according to the policies and procedures of the IRB of record

A protocol deviation may be considered an instance of Reportable New Information (RNI) if it:

- Has harmed or increased the risk of harm to one or more research participants.
- Has compromised the rights and welfare of the research subject
- Has damaged the scientific integrity of the data collected for the study.
- Results from willful or knowing misconduct on the part of the investigator(s).
- Demonstrates serious or continuing noncompliance with federal regulations, State laws, or University policies.

All protocol deviations will be documented by the study team on a paper or electronic deviation tracking log (see [NOTIS](#) for copy of log) as they occur. The deviation tracking log must be made available upon request for review by the assigned QAM. The deviation tracking log must be reviewed, signed, and dated by the investigator prior to each monitoring visit, or otherwise in a timely manner, whichever occurs first. The PI signed and dated deviation tracking log will be uploaded to eCRFs prior to each scheduled monitoring visit or upon request.

Deviations will be reviewed per the [DSMP](#).

10.8 Investigator Obligations

The Principal Investigator is responsible for the conduct of the clinical trial at the site in accordance with Title 21 of the Code of Federal Regulations and/or the Declaration of Helsinki. The PI is responsible for personally overseeing the treatment of all study patients. The PI is responsible for personally overseeing the treatment of all study patients. The PI must assure that all study site personnel, including sub-investigators and other study staff members, adhere to the study protocol and all FDA/GCP/NCI regulations and guidelines regarding clinical trials both during and after study completion.

The Principal Investigator at each institution or site will be responsible for assuring that all the required data will be collected, entered onto the appropriate eCRFs, and submitted within the study-specific timeframes. Periodically, monitoring visits may be conducted and the Principal Investigator will provide access to his/her original records to permit verification of proper entry of data. The study may also be subject to routine audits by the Audit Committee, as outlined in the [DSMP](#).

10.9 Publication Policy

All potential publications and/or data for potential publications (e.g. manuscripts, abstracts, posters, clinicaltrials.gov releases) must be approved in accordance with the DSMC Data Release Policies and Processes. The assigned QAM will prepare a preliminary data set for DSMC approval no later than 3 months after the study reaches its primary completion date, as defined by ClinicalTrials.gov. This is the date that the final patient was examined or received an intervention for the purposes of final collection of data for the primary outcome, whether the clinical trial concluded according to the pre-specified protocol or was terminated. If the investigator would like data release to be approved by the DSMC prior to when study design specifies, and/or prior to three months after a study's primary completion date, the PI must send a written request for data approval to the QAM which includes justification. Requests must be made a minimum of six to eight weeks in advance of the expected deadline. The request will be presented to the DSMC at their next available meeting. Any DSMC decisions regarding data release will be provided to the PI. If the request is approved, the QAM will present the data set to the DSMC for approval. A final, DSMC-approved dataset, as applicable, will then be released 6-8 weeks after the request was made. The investigators are expected to use only DSMC-approved data and statistical analyses any time they are disseminating trial data. The investigators must send a copy of the draft abstract/poster/manuscript to the study's biostatistician and assigned QAM to confirm that the DSMC-approved data and statistical analyses are used appropriately.

Once the biostatistician and QAM gives final approval, the publication may be submitted to external publisher.

11 REFERENCES

1. Smerage, J.B., et al., *Circulating tumor cells and response to chemotherapy in metastatic breast cancer: SWOG S0500*. J Clin Oncol, 2014. **32**(31): p. 3483-9.
2. Torre, L.A., et al., *Global cancer statistics, 2012*. CA Cancer J Clin, 2015. **65**(2): p. 87-108.
3. Metzger-Filho, O., et al., *Patterns of Recurrence and outcome according to breast cancer subtypes in lymph node-negative disease: results from international breast cancer study group trials VIII and IX*. J Clin Oncol, 2013. **31**(25): p. 3083-90.
4. Yonemori, K., et al., *Development and verification of a prediction model using serum tumor markers to predict the response to chemotherapy of patients with metastatic or recurrent breast cancer*. J Cancer Res Clin Oncol, 2008. **134**(11): p. 1199-206.
5. Marks, J.R., et al., *Construction and analysis of the NCI-EDRN breast cancer reference set for circulating markers of disease*. Cancer Epidemiol Biomarkers Prev, 2015. **24**(2): p. 435-41.
6. Cristofanilli, M., et al., *Circulating tumor cells, disease progression, and survival in metastatic breast cancer*. N Engl J Med, 2004. **351**(8): p. 781-91.
7. Cristofanilli, M., et al., *Circulating tumor cells: a novel prognostic factor for newly diagnosed metastatic breast cancer*. J Clin Oncol, 2005. **23**(7): p. 1420-30.
8. Giuliano, M., et al., *Circulating tumor cells as early predictors of metastatic spread in breast cancer patients with limited metastatic dissemination*. Breast Cancer Res, 2014. **16**(5): p. 440.
9. Nole, F., et al., *Variation of circulating tumor cell levels during treatment of metastatic breast cancer: prognostic and therapeutic implications*. Ann Oncol, 2008. **19**(5): p. 891-7.
10. Bidard, F.C., et al., *Clinical validity of circulating tumour cells in patients with metastatic breast cancer: a pooled analysis of individual patient data*. Lancet Oncol, 2014. **15**(4): p. 406-14.
11. Munzone, E., et al., *Prognostic value of circulating tumor cells according to immunohistochemically defined molecular subtypes in advanced breast cancer*. Clin Breast Cancer, 2012. **12**(5): p. 340-6.
12. Giuliano, M., et al., *Circulating tumor cells as prognostic and predictive markers in metastatic breast cancer patients receiving first-line systemic treatment*. Breast Cancer Res, 2011. **13**(3): p. R67.
13. Aktas, B., et al., *Comparison of estrogen and progesterone receptor status of circulating tumor cells and the primary tumor in metastatic breast cancer patients*. Gynecol Oncol, 2011. **122**(2): p. 356-60.
14. Fehm, T., et al., *Detection and characterization of circulating tumor cells in blood of primary breast cancer patients by RT-PCR and comparison to status of bone marrow disseminated cells*. Breast Cancer Res, 2009. **11**(4): p. R59.
15. Kurbel, S., *Selective reduction of estrogen receptor (ER) positive breast cancer occurrence by estrogen receptor modulators supports etiological distinction between ER positive and ER negative breast cancers*. Med Hypotheses, 2005. **64**(6): p. 1182-7.
16. Esserman, L.J., et al., *Tamoxifen may prevent both ER+ and ER- breast cancers and select for ER- carcinogenesis: an alternative hypothesis*. Breast Cancer Res, 2005. **7**(6): p. R1153-8.
17. Garcia-Becerra, R., et al., *Mechanisms of resistance to endocrine therapy in breast cancer: focus on signaling pathways, miRNAs and genetically based resistance*. Int J Mol Sci, 2012. **14**(1): p. 108-45.
18. Babayan, A., et al., *Heterogeneity of estrogen receptor expression in circulating tumor cells from metastatic breast cancer patients*. PLoS One, 2013. **8**(9): p. e75038.
19. Sirohi, B., et al., *Platinum-based chemotherapy in triple-negative breast cancer*. Ann Oncol, 2008. **19**(11): p. 1847-52.
20. Isakoff, S.J., et al., *TBCRC009: A Multicenter Phase II Clinical Trial of Platinum Monotherapy With Biomarker Assessment in Metastatic Triple-Negative Breast Cancer*. J Clin Oncol, 2015. **33**(17): p. 1902-9.
21. Tutt A, E.P., Kilburn L, et al. . *The TNT trial: A randomized phase III trial of carboplatin (C) compared with docetaxel (D) for patients with metastatic or recurrent locally advanced triple negative or BRCA1/2 breast cancer (CRUK/07/012)*. in AACR/SABC. 2015.

22. Nishino, M., et al., *Developing a common language for tumor response to immunotherapy: immune-related response criteria using unidimensional measurements*. Clin Cancer Res, 2013. **19**(14): p. 3936-43.
23. Russnes, H.G., et al., *Insight into the heterogeneity of breast cancer through next-generation sequencing*. J Clin Invest, 2011. **121**(10): p. 3810-8.
24. Badve, S. and H. Nakshatri, *Breast-cancer stem cells-beyond semantics*. Lancet Oncol, 2012. **13**(1): p. e43-8.
25. Meloni, F., et al., *Foxp3 expressing CD4+ CD25+ and CD8+CD28- T regulatory cells in the peripheral blood of patients with lung cancer and pleural mesothelioma*. Hum Immunol, 2006. **67**(1-2): p. 1-12.
26. Song, G., et al., *Elevated level of peripheral CD8(+)CD28(-) T lymphocytes are an independent predictor of progression-free survival in patients with metastatic breast cancer during the course of chemotherapy*. Cancer Immunol Immunother, 2013. **62**(6): p. 1123-30.
27. Zhang, L., et al., *Differential impairment of regulatory T cells rather than effector T cells by paclitaxel-based chemotherapy*. Clin Immunol, 2008. **129**(2): p. 219-29.
28. Traverso, I., et al., *Cyclophosphamide inhibits the generation and function of CD8(+) regulatory T cells*. Hum Immunol, 2012. **73**(3): p. 207-13.
29. Lutsiak, M.E., et al., *Inhibition of CD4(+)25+ T regulatory cell function implicated in enhanced immune response by low-dose cyclophosphamide*. Blood, 2005. **105**(7): p. 2862-8.
30. Tohyama, N., et al., *Influence of anticancer agents on cell survival, proliferation, and CD4+CD25+Foxp3+ regulatory T cell-frequency in human peripheral-blood mononuclear cells activated by T cell-mitogen*. Int Immunopharmacol, 2013. **15**(1): p. 160-6.
31. Chen, C., et al., *Suppressive effects of gemcitabine plus cisplatin chemotherapy on regulatory T cells in nonsmall-cell lung cancer*. J Int Med Res, 2015. **43**(2): p. 180-7.
32. Zamarin, D. and M.A. Postow, *Immune checkpoint modulation: rational design of combination strategies*. Pharmacol Ther, 2015. **150**: p. 23-32.
33. Lake, R.A. and B.W. Robinson, *Immunotherapy and chemotherapy--a practical partnership*. Nat Rev Cancer, 2005. **5**(5): p. 397-405.
34. Adams, D.L., et al., *Circulating giant macrophages as a potential biomarker of solid tumors*. Proc Natl Acad Sci U S A, 2014. **111**(9): p. 3514-9.
35. Al-Kateb, H., et al., *Identification of major factors associated with failed clinical molecular oncology testing performed by next generation sequencing (NGS)*. Mol Oncol, 2015. **9**(9): p. 1737-43.
36. Cooper, Z.A., et al., *BRAF inhibition is associated with increased clonality in tumor-infiltrating lymphocytes*. Oncoimmunology, 2013. **2**(10): p. e26615.
37. Tumei, P.C., et al., *PD-1 blockade induces responses by inhibiting adaptive immune resistance*. Nature, 2014. **515**(7528): p. 568-71.
38. Robert, L., et al., *CTLA4 blockade broadens the peripheral T-cell receptor repertoire*. Clin Cancer Res, 2014. **20**(9): p. 2424-32.
39. Robins, H.S., et al., *Comprehensive assessment of T-cell receptor beta-chain diversity in alpha beta T cells*. Blood, 2009. **114**(19): p. 4099-4107.
40. Carlson, C.S., et al., *Using synthetic templates to design an unbiased multiplex PCR assay*. Nature Communications, 2013. **4**.

12 APPENDICES

12.1 ECOG Performance Status

Grade	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.
1	Symptoms, but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work).
2	In bed <50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	In bed >50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Dead.
* As published in Am. J. Clin. Oncol.: Oken, M.M., Creech, R.H., Tormey, D.C., Horton, J., Davis, T.E., McFadden, E.T., Carbone, P.P.: Toxicity And Response Criteria Of The Eastern Cooperative Oncology Group. Am J Clin Oncol 5:649-655, 1982. The Eastern Cooperative Oncology Group, Robert Comis M.D., Group Chair.	

12.2 Common Terminology Criteria for Adverse Events V4.03 (CTCAE)

The descriptions and grading scales found in the revised NCI Common Terminology Criteria for Adverse Events (CTCAE) version 4.03 will be utilized for AE reporting. (<http://ctep.cancer.gov/reporting/ctc.html>)

12.3 Response Evaluation Criteria in Solid Tumors (RECIST) 1.1

Criteria for Evaluating Response in Solid Tumors

RECIST version 1.1* will be used in this study for assessment of tumor response. While either CT or MRI may be utilized, as per RECIST 1.1, CT is the preferred imaging technique in this study.

* As published in the European Journal of Cancer:

E.A. Eisenhauer, P. Therasse, J. Bogaerts, L.H. Schwartz, D. Sargent, R. Ford, J. Dancey, S. Arbuck, S. Gwyther, M. Mooney, L. Rubinstein, L. Shankar, L. Dodd, R. Kaplan, D. Lacombe, J. Verweij. New response evaluation criteria in solid tumors: Revised RECIST guideline (version 1.1). Eur J Cancer. 2009 Jan;45(2):228-47.

In addition, volumetric analysis will be explored by central review for response assessment.

12.4 Summary of Changes

IRB & FDA Requested Changes – June 19, 2017			
Sections(s) Affected	Prior Version	Requested Changes	Rationale
List of Abbreviations; Section 4.3.2 Carboplatin Dose Delays & Modifications	n/a	Added “QAM” to List	Per FDA, clarification of “QAM”, which stands for Quality Assurance Monitor
Study Summary	Study Duration: 36 months	Study Duration: 60 months	Per FDA, total study duration was corrected to include 18 months of accrual, 24 months of treatment, and 18 months of follow up. Therefore, total time of the study would be 60 months.
Section 1.2.2 Rationale for Dose Selection/Regimen/Mo dification	n/a	Explanation of why carboplatin was chosen as a chemotherapy agent	Per FDA, included justification for why carboplatin is used for this study.
Section 3.2 Exclusion Criteria	“Histologically or cytologically confirmed HER2-positive (3+ by IHC or non-amplified by FISH) according to ASCO/CAP guidelines”	“Histologically or cytologically confirmed HER2-positive (3+ by IHC or amplified by FISH) according to ASCO/CAP guidelines”	Per FDA, “non-amplified” was written in error and has been corrected to “amplified”.
Section 4.3.2 Carboplatin Dose Delays & Modifications	n/a	Clarification of dose level of carboplatin when dose is reduced.	Per FDA, clarification of dose reduction for carboplatin.
	Carboplatin dose reduction beyond -1.	Carboplatin will be discontinued beyond -1 dose reduction.	Only 1 dose reduction will be allowed, beyond that treatment with carboplatin will be discontinued.
Section 4.8 Duration of Therapy	n/a	Clarification of duration of therapy	Per IRB, clarification of the duration of therapy. Patients will receive treatment for 24 months or 35 administrations (whichever is later) until any of the listed scenarios occur.
Section 7.1.12 Pembrolizumab Nursing Implications	“Pembrolizumab will be administered first, followed sequentially by paclitaxel and carboplatin ”	“Pembrolizumab will be administered first, followed sequentially by carboplatin”	Per FDA, paclitaxel was written in error and has been removed.
Section 7.2.5 Carboplatin Protocol Dose Specifics	“Carboplatin will be given by IV at an area under the	“Carboplatin will be given by IV at an area under the	Per FDA, “dose of 2” was written in error and has been corrected to be “dose of 5”.

	curve (AUC) dose of 2. "	curve (AUC) dose of 5. "	
Section 9 Correlatives/Special Studies	In correlative sample table, tissue collection was listed as mandatory	In correlative sample table, tissue collection is listed as optional	Per FDA, archival tissue is requested in the Inclusion criteria, however lack of archival tissue does not make the patient ineligible for the study.
Amendment 1 – October 17, 2017			
Sections(s) Affected	Prior Version	Requested Changes	Rationale
Cover Page	Cesar Augusto Santa-Maria, MD and Sarika Jain, MD listed as Sub-Investigators	Cesar Augusto Santa-Maria, MD and Sarika Jain, MD removed as Sub-Investigators	Physicians are no longer at Northwestern University
Study Schema; Study Summary; Section 4.1 treatment overview; Section 4.7 Withdrawal/Discontinuation	n/a	Added "or 35 administrations of study medication, whichever is later"	Previous version of study schema indicated treatment was 24 months, however it is 24 months or 35 administrations of study medication, whichever is later
Study Summary	Short Title: Pembrolizumab and carboplatin in CTC+ HER2-MBC	Short title removed	Administrative. Short title no longer required in NU IIT template.
	n/a	In Objectives, included "and patients with CTC positive, TNBC, MBC not exposed to other treatments in the metastatic setting."	Objective statement excluded part of the eligible patient population in error.
Section 2.2 and Section 6.4 Secondary Objectives and Endpoints	Previous list of objectives	Added one more objective for toxicity and safety assessment using NCI CTCAE 4.03	Since it is a Phase II study, a toxicity objective and endpoint is essential
Section 3.1.1 Inclusion Criteria	n/a	Included that patients "should have been exposed to anthracyclines and taxane in neoadjuvant and adjuvant settings (first-line)"	Inclusion criteria updated to include the requirement that patients should have been exposed to anthracyclines and taxane in neoadjuvant and adjuvant settings (first-line).

Section 3.1.7 (Inclusion Criteria)	Allows patients with liver metastases to have <5 xULN criteria for AST and ALT	Removed provision allowing patients with liver metastases to have <5 xULN criteria for AST and ALT.	Changes to address previous SAEs that have occurred in patients while on study
Section 3.1.9 Inclusion Criteria; Section 3.1.10 Inclusion Criteria	Adequate method of contraception as outlined in the appendices	Adequate method of contraception as outlined in Section 4.5	Readers are now directed to section 4.5 instead of the appendices for adequate methods of contraception.
Section 3.1.11 Inclusion Criteria	n/a	Included that patients must be willing to “comply with all study procedures”	Included language from NU IIT template.
Section 3.2.1 Exclusion Criteria	n/a	Included a note “A HER2 result of 2+ by ICH is equivocal and requires a reflex test (same specimen using the alternative test) or new test (new specimen, if available, using same or alternative test)”	Added text to clarify the result of HER2 2+ by IHC result.
Section 3.2.2 Exclusion Criteria	“< or equal to 28 days”	“≤ 28 days”	For clarification.
Section 3.2.3 Exclusion Criteria	n/a	Added criteria that patients with prior carboplatin in the metastatic setting are NOT eligible, but prior carboplatin as neoadjuvant or adjuvant treatment are permitted	For safety, specificity and clarity
Section 3.2.9 Exclusion Criteria	Criteria stating that subjects undergoing major surgery should recover adequately before starting study drug.	Added a wash-out period of 2 weeks along with adequate recovery prior to starting study drug.	For safety and clarity

Section 4.3.1 (Pembrolizumab Dose Delay & Modifications), Section 4.5 (Rescue Medications and Supportive Care)	Dose modifications for pembrolizumab immune-related adverse events	Included updated table for pembrolizumab immune-related adverse events.	New pembrolizumab dose modification table provided by Merck.
Section 4.4.1	Language regarding recording of Concomitant medications as Events of Clinical Interest.	Added that the NU SAE form will be used for reporting these	For clarity
Section 4.4.2 Prohibited Concomitant Medications	n/a	Under Radiation Therapy, included the note that “radiated lesions cannot be used as a target lesion for disease assessment”	For clarity: Included text to specify that radiated lesions cannot be used as a target lesion for disease assessment.
Section 4.9.3	Survival follow-up description	Added language that “Information regarding the subject’s next line of therapy and clinical response will be collected if a new treatment is initiated while the patient is on follow up.”	To facilitate collection of vital information regarding post PD treatment and response
Section 5.1 Study Flow Chart	“Discon”	“Discontinuation”	For clarification.
	“End of Treatment”	“EOT” added to abbreviation list	For clarification.
	Previous information	Lesion Photographs was added. Included footnote #7 “Optional photos for disease assessments may be taken by study	For safety, clarity and flexibility

		staff on a study-supplied digital camera per treating physician discretion. It is preferred that these photos be taken on a white background (wall of patient room) using a ruler measurement when possible. Photographs should not include any identifying characteristics.” Footnote 4: added language to match section 4.9.3 Footnote 8: FOCBP should have a negative urine or serum pregnancy test within 7 days prior to registration; and this must be repeated within 3 days (72 hours) prior to first dose of study drug.	
Section 5.2.1 Trial Procedures	Under Vital Signs, “Vital signs should include temperature, pulse, respiratory rate , weight and blood pressure”	Removed “respiratory rate” as part of vital signs	Collection of respiratory rates has been removed from protocols due to the frequent deviations.
	Under ECOG Performance Status, directed readers to Appendices 11.1	Under ECOG Performance Status, directed readers to Appendices 12.1	Readers were earlier directed to the incorrect section. Error corrected.

Section 6.7.3 Table 6: irRECIST	Details for Tumor Imaging and Treatment after 1st Radiologic Evidence of PD or SD, CR or PR	<i>Added Note: Clinical stability / instability will be a clinical decision and this should be documented in the patient's chart with proper supporting documents.</i>	For clarity and compliance purposes
7.2.5 Protocol Dose Specifics (Carboplatin)	n/a	Added "Carboplatin dose calculation is to be per institutional standard and based on the serum creatinine of the day."	For safety and clarity
	NOTE: When using the Calvert equation, GFR should not exceed 125 mL/min. Thus, the maximum carboplatin dose is 2 x (125 + 25), or 300 mg.	This note was removed	
Section 7.2.9 Availability and supply of Carboplatin	Previous details	Added that commercial supply of carboplatin will be used in this study	For clarity

Section 8.2.1 Adverse Event	n/a	Included text “If patient experiences PD, they will come off study treatment. If patient experiences a SAE from drug toxicity follow toxicity management guidelines and/or physician discretion.”	For clarification: that progression of the cancer under study is not considered an AE. However, If patient experiences PD, they will come off study treatment, but will still be followed up . If patient experiences a SAE from drug toxicity, toxicity management guidelines are to be followed and/or physician discretion.
Section 9 Correlative/Special Studies	Correlative sample table	Correlative Sample table was removed Included short description of correlative samples “Blood and tissue will be collected as part of study procedures (see Section 5). Details on sample collection, processing, storage and shipment can be found in the Lab Manual.”	Correlative sample table is typically removed prior to IRB submission. This study has a separate laboratory manual with all the correlative details This section includes brief description of correlative studies for a quick references and also refers readers to the lab manual.
Section 10.2 Amendments	“[insert sponsor name if applicable]”	“Merck Sharp & Dohme Corp”	Administrative.
Section 12.2 Appendices	“Common Terminology Criteria for Adverse Events V4.0”	“Common Terminology Criteria for Adverse Events V4.03”	Administrative.
Amendment 2 – July 30, 2018			
Sections(s) Affected	Prior Version	Changes made	Rationale
Cover page	N/a	Added Ami Shah, MD as Sub-Investigator	Administrative change
Study Schema; Study Summary; Section 1.2.2 (Rationale for Dose Selection/Regimen/Modification); Section	Included HR+Her2-endocrine refractory patient population	Removed HR+Her2-endocrine refractory patient population as well as the related	Exclude endocrine-refractory disease because these patients are too sick to be eligible for endpoint assessment. There are other studies available for this population.

1.2.3 (Rationale for Efficacy Endpoints); Section 1.3 (Rationale for Current Study); Section 2.1 (Primary Objectives); Section 2.2 (Secondary Objectives); Section 3 (Patient Eligibility); Section 4.11 (Patient Replacement); Section 6.3 (Primary Endpoint); Section 6.4 (Secondary Endpoints); Section 6.5 (Exploratory Endpoints)		background information	
Study Schema; Study Summary; 1.2.2 (Rationale for Dose Selection/Regimen/Modification); Section 4.2 (Treatment Administration); Section 4.2.2 (Carboplatin); Section 5.1 (Study Flow Chart);	Carboplatin treatment begins at Cycle 3	Carboplatin treatment begins with pembrolizumab treatment at Cycle 1	Change in treatment plan to expose patients to carboplatin treatment earlier
Section 3.2.18 (Exclusion Criteria); Section 3.2.20 (Exclusion Criteria)	n/a	Exclusion criteria updated to match the Merck Template.	Updated specific criteria to align with the Merck Template.
Section 5.1 (Study Flow Chart)	n/a	Footnote 13 edited to clarify tumor imaging requirements	Clarify requirements for tumor imaging
Section 6.3 (Primary Endpoint); Section 6.4 (Secondary Endpoints); Section 6.5 (Exploratory Endpoints)	Any patient who has completed at least 2 cycles of study treatment are evaluable for this endpoint.	Any patient who has completed at least 3 cycles of study treatment are evaluable for this endpoint.	Patients will need to complete 3 cycles of treatment (at the time of imaging) to be eligible for endpoint assessment.
Section 6.6 (Statistical Analysis)	n/a	Added "Prior to Amendment 2 (dated July 30, 2018), the first six patients who were treated with sequential pembrolizumab followed by combination therapy and included endocrine-refractory disease	Clarification requested by the Scientific Review Committee. Explanation of how the data from the first six patients will be used since the amendment changed the way the treatment was administered and it changed the type of patients in the study.

		will be used only to assess safety. This has been completed and there were no dose limiting toxicities in these patients. As the protocol is written, 38 patients will be accrued in order to obtain 32 patients who have been treated and are evaluable for progression-free survival (PFS). These numbers will be applied to the amended patient population. The null hypothesis will still be rejected if there is evidence that the median PFS exceeds 4.9 months."	
Section 8.3.5 (Reporting to the FDA)	"DMC"	"DSMC"	Administrative update
Amendment 3 – August 6, 2020			
Sections(s) Affected	Prior Version	Changes made	Rationale
Title page	Alfred Rademaker was listed as the statistician	Removed Alfred Rademaker and added Borko Jovanovic, PhD as statistician.	Assignment of new statistician since Alfred Rademaker has retired from NU
Throughout	Patients with CTC positive, triple negative (TN) MBC previously treated with anthracyclines and taxanes in primary setting	Addition of patients in the de-novo stage IV category as well. Hence, now it reads as : "patients with CTC positive, triple negative (TN) MBC previously treated with anthracyclines and taxanes in primary setting or de-novo stage IV.	To facilitate recruitment <i>The rationale being that TNBC either recurrent or de-novo with higher CTCs identify a very aggressive cohort with intrinsic chemotherapy resistance.</i>

Section 3.2.4 and 3.2.11 Exclusion criteria an	Language regarding use of steroids in different scenarios that make patients ineligible for study	Added note to clarify that steroids as premedication are permitted	<i>For increased clarity and consistency</i>
Eligibility Criteria Sections 3.2.20; 3.1.8 3.1.9	Section 3.2.20 had exclusion criteria for Hep B and C “Has a known history of Hepatitis B (defined as Hepatitis B surface antigen [HBsAg] reactive) or known active Hepatitis C virus (defined as HCV RNA [qualitative] is detected) infection.”	This criteria has been removed and reworded as two inclusion criteria 3.1.9 and 3.1.9 . Language is in accordance with current NU template and in alignment with CTEP recommendations	<i>For increased clarity and to align with current standard template language</i>
Section 4.2 Table 2 Trial treatment	Table outlining trial treatment with details	Added footnote for carboplatin: “steroids as pre-medication for carboplatin is permitted”	<i>For clarity and consistency</i>
Section 4.4.2 Prohibited Concomitant Medications	Language detailing prohibition of steroids. PI to be consulted for use of physiologic doses of steroids	Added note to clarify that: “steroids as pre-medication are permitted in this study”	<i>For clarity and to maintain consistency with updates made to the rest of the protocol</i>
Section 5 Study procedures table Footnote 13	Tumor imaging will occur every 9 weeks following initiation of treatment. CT (chest/abdomen/p	Reworded to state: “CT c/a/p AND bone scan. In place of bone scan, a PET/CT may be used. If CT is contraindicated, MRI may be used. Any	<i>For increased clarity and convenience</i>

	elvis) and either 1) bone scan or 2) PET/CT at the discretion of treating physician are allowed.	additional imaging such as MRI can be completed at the physician's discretion. The same modality should be used throughout the study. <i>Note: Bone scans/PET are to be repeated at each time point ONLY if the patient has bone metastasis or if it is clinically indicated. If the original bone scan/PET during screening is negative, then bone scans/PET are not required in subsequent visits. "</i>	
Section 6.7.3 Table 7 irRECIST	This table was numbered as Table 6 in error Language for when repeat scan shows SD,PR or CR, in the Clinically stable category, treatment directions were "Continue study treatment at the Investigator's discretion."	Table numbering updated to Table 7 In the same category added language to clarify that "Next scan should occur according to the every 9 week (63 ± 7 days) imaging schedule"	<i>For increased clarity</i>

Section 7.1.9 Availability and Supply Of pembrolizumab	Contact for drug ordering and other details	Updated contact information and drug ordering details	Administrative Based on information received from Merck
Section 4.1	Carboplatin is still stated to begin at Cycle 3. This section was not updated with the rest of the protocol in an earlier amendment	Corrected language to state that carboplatin will begin at Cycle 1 along with pembrolizumab	Correction of error [to maintain consistency]
Amendment 4 – December 17, 2020			
Sections(s) Affected	Prior Version	Changes made	Rationale
Study schema	First line therapy or denovo stage IV	Removed this sentence	For consistency with new modifications made to the eligibility section of the protocol

Section 3.1 and 3.1.1 Inclusion criteria And study schema	<i>Patients must be Hormone receptor (HR) negative and HER-2 negative (TNBC) metastatic breast cancer and have not received prior chemotherapy for metastatic disease and should have been exposed to anthracyclines and taxane in neoadjuvant and adjuvant settings (first-line) or be de-novo stage IV</i>	<i>Modified this criteria by removing “should have been exposed to anthracyclines and taxane in neoadjuvant and adjuvant settings (first-line)”</i> Added a new 3.1.2 criteria to state that “Patients should be either Stage IV de-novo or develop recurrence after completion of neoadjuvant/adjuvant therapy.”	<i>For flexibility in order to facilitate enrollment</i>
Section 3.2.3 Exclusion criteria	Patients who had prior carboplatin in the metastatic setting are not eligible. This criteria contains a note which implies that <i>patients could have been exposed to carboplatin in neoadjuvant/adjuvant setting;</i>	The note has been updated to state that “Patients who have been exposed to carboplatin in neoadjuvant or adjuvant setting will be allowed to enroll, if they have progressed $\geq$ 6 months from completion of treatment.”	<i>For clarity</i>

Section 2 And Section 6.4 Objectives and endpoints And Study Schema	Target population for this study was stated as patients with CTC positive, triple negative (TN) MBC previously treated with anthracyclines and taxanes in primary setting or de-novo stage IV	Study population description has been updated to “patients with CTC positive, triple negative (TN) MBC who have not received prior chemotherapy for metastatic disease or have developed recurrence after completion of neoadjuvant/adjuv ant therapy or are de-novo stage IV”	<i>To align with updates made to the eligibility criteria</i>
Amendment 5 – April 19, 2021			
<i>Sections(s) Affected</i>	<i>Prior Version</i>	<i>Changes made</i>	<i>Rationale</i>
Title page, Study summary and Section 3 overview	It was a single center study with the Northwestern university Robert H. Lurie Cancer Center, Chicago, as the only site	Now its is a multi- center study. Affiliate site is Northwestern Memorial Healthcare with Perry Manini, DO as the Affiliate PI. They will be opening the study at the Delnor, Warrenville and Kishwaukee campuses.	<i>On boarding of Affiliate site for the study.</i>
	The study PI, Dr.Cristofanilli's email address was his NM email ID	Dr.Cristofanilli's email address has been updated to his Northwestern email ID	<i>Administrative</i>
Section 3.2.13 Exclusion criteria	Patients who have evidence of active, noninfectious pneumonitis or have a history of severe pneumonitis that required treatment with steroids are not eligible for this study.	The term interstitial lung disease has been added with noninfectious pneumonitis	<i>As mandated by Merck based on updated Merck template v12.1</i>

Section 3.2.20 Exclusion criteria	Has received a live vaccine within 30 days of planned start of study therapy.	Criteria updated to state : <i>Has received a live vaccine or live attenuated vaccine within 30 days of planned start of study therapy.</i> <i>Administration of killed vaccines is allowed.</i>	<i>As mandated by Merck based on updated Merck template v12.1</i>
Section 10 Study management	Previous NU template language	Updated with current NU template language. A section with instructions for participating sites Has been inserted since this is now a multi-center study Sub-sections have been re-numbered accordingly to accommodate this new section	<i>Administrative</i>
<i>Amendment 6 dated 11.30.2021</i>			
<i>Sections(s) Affected</i>	<i>Prior Version</i>	<i>Changes made</i>	<i>Rationale</i>
Title page And study schema	Massimo Cristofanilli, MD as PI and IND holder	Removed Dr.Cristofanilli, MD as PI and IND holder and added Dr.Lisa E.Flaum, MD as PI and IND holder. All appropriate contact information inserted. Removed Dr.Lisa Flaum from the Sub-I list	<i>Dr.Cristofanilli is leaving Northwestern University and is handing over the study to Dr. Lisa E.Flaum</i>

	Affiliate sites were inserted in the title page in the last amendment [Northwestern Memorial Healthcare (Delnor, Warrenville and Kishwaukee campuses)]	Removed affiliate sites	<i>Dr.Cristofanilli decided to close the study at the recommendation of the SRC (due to low accrual numbers). The study was closed before it was launched at the affiliate sites.</i>
Section 3.0	Massimo Cristofanilli, MD listed as PI along with his phone number for recruitment questions Study was changed to multi-center when affiliate sites were added	Removed Dr.Cristofanilli, MD as PI and added Dr.Lisa E. Flaum, MD as PI and with contact phone number. Study changed to single center again as affiliate sites have been removed in this amendment	<i>Dr.Cristofanilli is leaving Northwestern University and is handing over the study to Dr.Lisa E.Flaum</i> <i>[Note: This change is being done for consistency. Currently, the study is closed to accrual]</i>